MMS236 - Aircraft Design

Lecture 3

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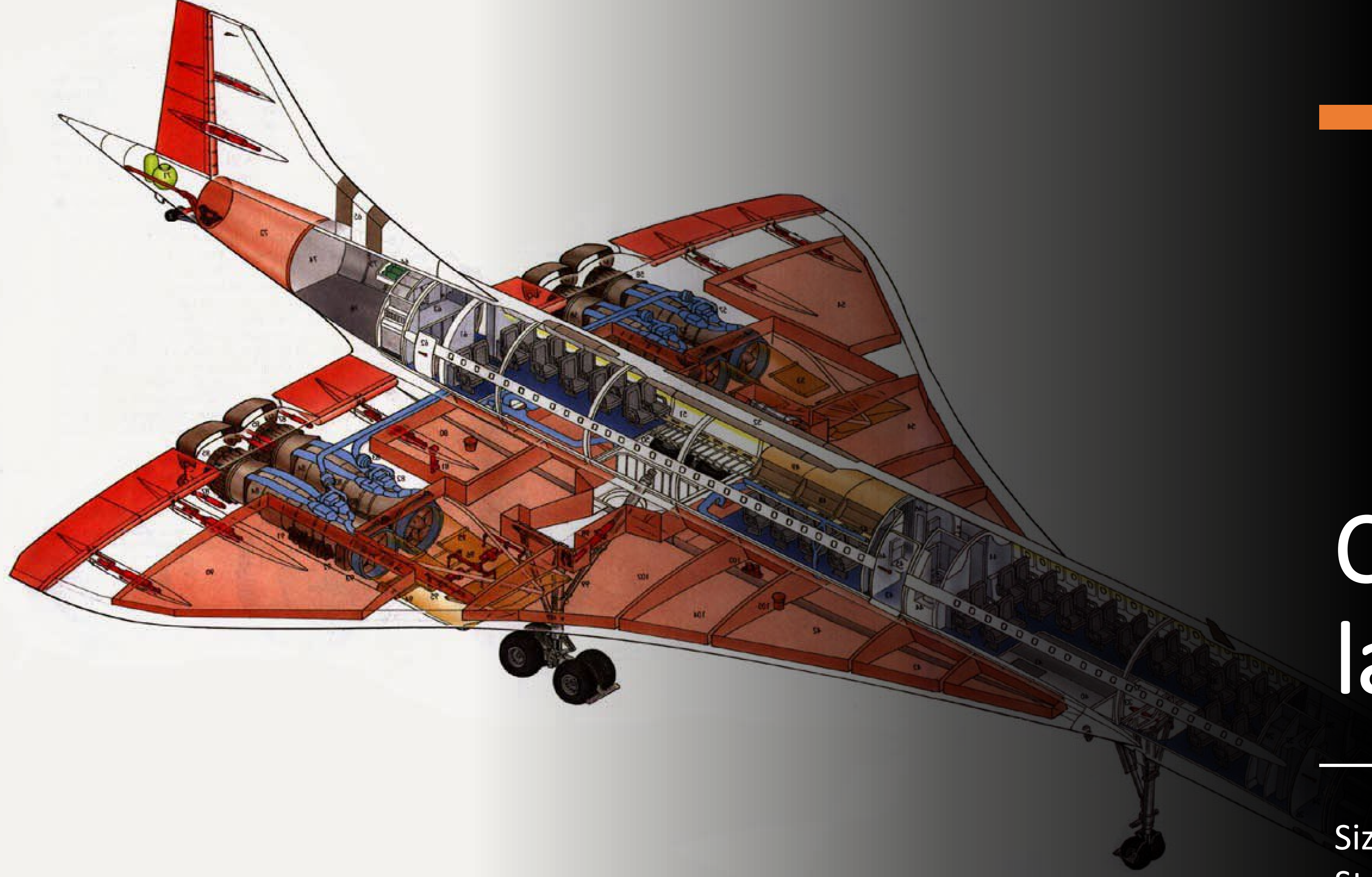
UNIVERSITY OF TECHNOLOGY



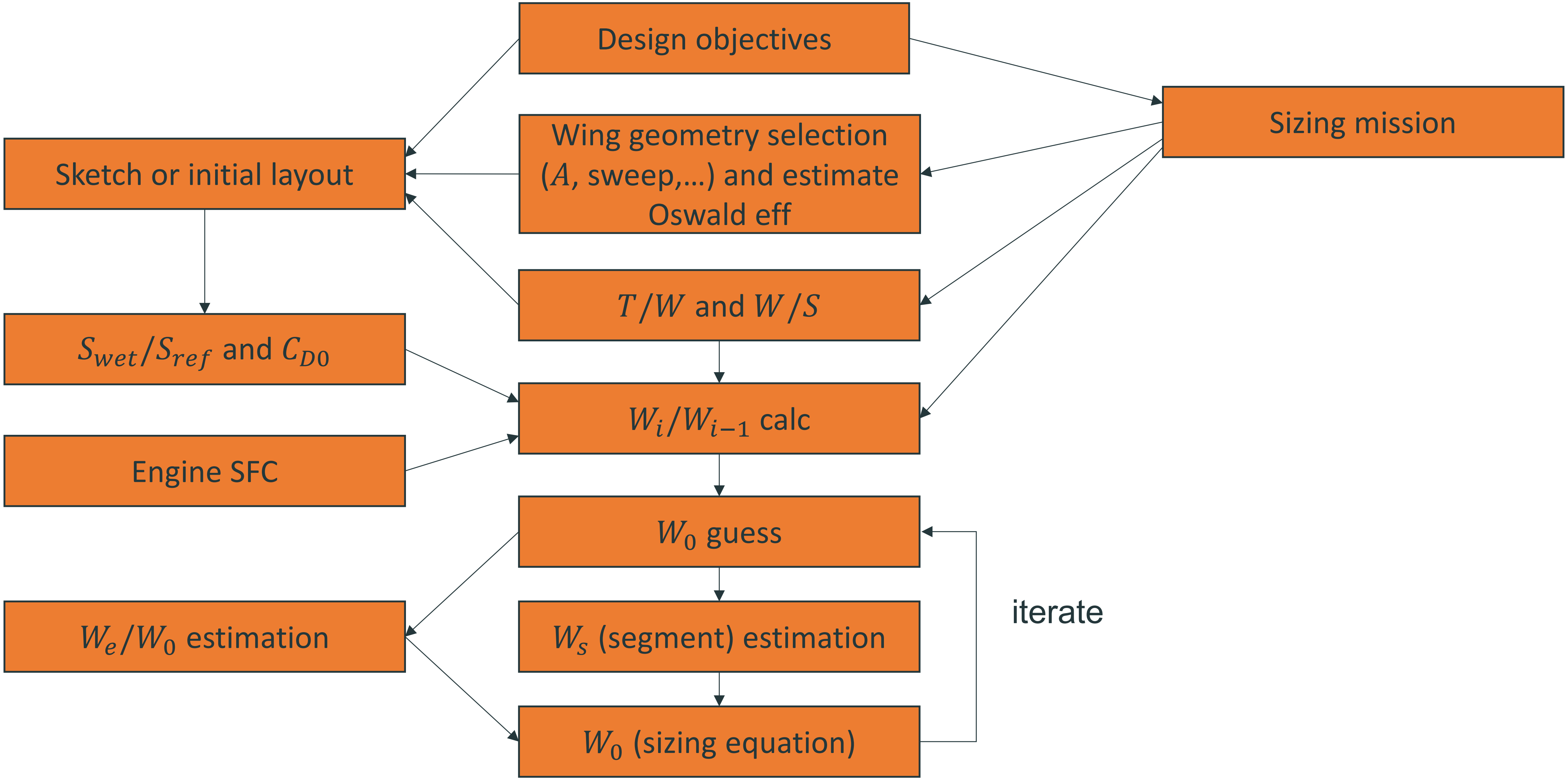
Aérospatiale/BAC Concorde

Conceptual layout

Sizing (continuation). Aerodynamic and Structural Considerations. Crew station, passengers and payload, landing gear a subsystems

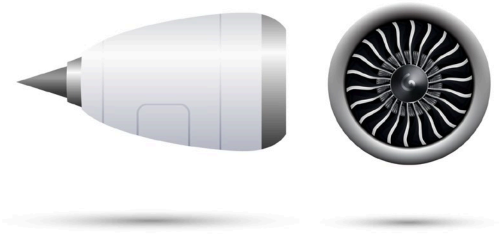
Initial sizing
(better method)

Sizing method



Initial sizing (improved method in Raymer)

- Steps through the mission to find fuel burn for each segment.
- Allows for payload drop.
- Can be used to size while considering a ”fixed” (existing) engine or ”rubber” (new) engine.



Initial sizing – Rubber engine

Remember the sizing equation?

- $W_0 = W_{crew} + W_{payload} + W_{dropped\ payload} + W_f + W_e$

  Or

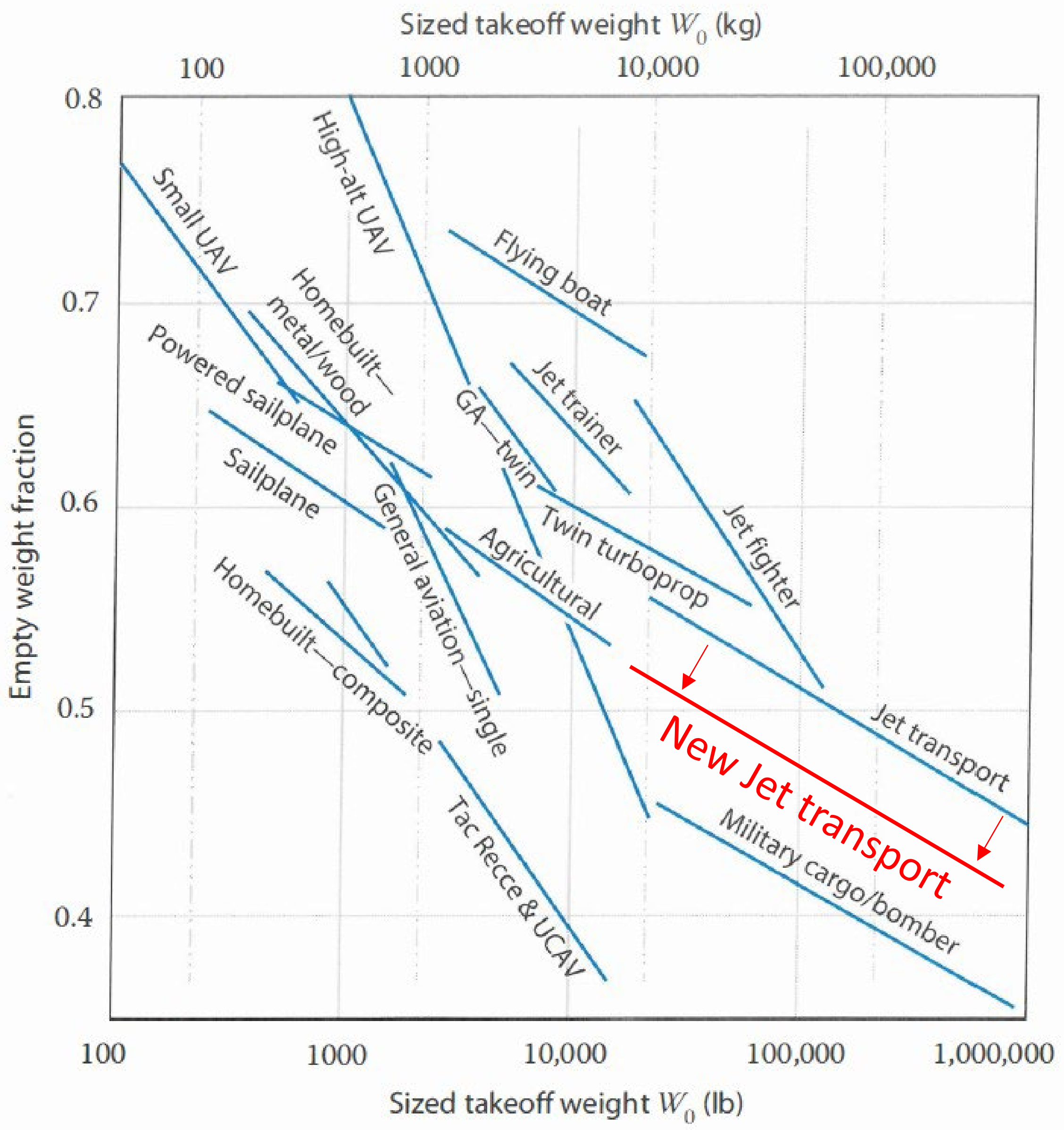
- $W_0 = W_{crew} + W_{payload} + W_{dropped\ payload} + W_f + \frac{W_e}{W_0} W_0$

- $\frac{W_e}{W_0}$ - predicted using statistical methods at this point - *we have not designed the aircraft!*

Empty weight fraction

Best approach

- Gather as much data as possible on similar aircraft (the more recent the better).
- Create your own $\frac{W_e}{W_0}$ vs. W_0 correlation.
  - remember that $\frac{W_e}{W_0}$ normally goes down with increasing W_0.
- Apply fudge factors as needed.



Empty weight fraction

Alternative (more time consuming)

- Use more independent variables (A, W/S, T/W, M) Tab. 6.1 and 6.2 in Raymer.

$$\frac{W_e}{W_0} = \left[a + b\, W_0^{C1} A^{C2} \left(\frac{T}{W_0}\right)^{C3} \left(\frac{W_0}{S}\right)^{C4} M_{\max}^{C5}\right] \cdot K_{vs}$$

fps units	a	b	C1	C2	C3	C4	C5
Jet fighter	-0.02	2.16	-0.1	0.2	0.04	-0.1	0.08
Military Cargo	0.07	1.71	-0.1	0.1	0.06	-0.1	0.05
Jet Transport	0.32	0.66	-0.13	0.30	0.06	-0.05	0.05

K_{vs} - variable sweep constant: 1.04 for variable sweep wings; 1.0 otherwise.

Note: Don’t forget to convert from lbs to kg.

Fuel weight

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- We now assume the possibility of payload drop
- The segment fuel burn is:

$$W_{f_i} = \left(1 - \frac{W_i}{W_{i-1}}\right) W_{i-1}$$

$\frac{W_i}{W_{i-1}}$ - mission segment weight fractions

1. Start by W_0 (guess) = W_1
2. Apply mission-segment weight fraction to calculate W_{f1}
3. $W_2 = W_1 - W_{f1} - W_{dropp}$ (if any)
4. Repeat

- The mission fuel burn is (including 5% contingency and 1% trapped fuel):

$$W_f = 1.06 \sum_{1}^{i} W_{f_i}$$

Weight fraction (W_i/W_{i-1}) – Cruise

$$\left(\frac{W_i}{W_{i-1}}\right)_{cruise} = \exp\left(\frac{-R\cdot(SFC)_{\mathrm{cruise}}\cdot g_0}{V\left(\frac{L}{D}\right)_{\mathrm{cruise}}}\right)$$

Where in cruise Lift = Weight, hence:

$$\left(\frac{L}{D}\right)_{\mathrm{cruise}} = \frac{C_L}{C_D} = \frac{C_L}{\underbrace{C_{D_0}}_{\text{Parasite}} + \underbrace{\underbrace{C_{Di}}_{\text{Induced}} + \underbrace{C_{D_{\mathrm{Lift-viscous}}}}_{\text{Lift dependent viscous}} + \underbrace{C_{D\,wave}}_{\text{Compressibility drag}}}_{\text{Lift dependent}}}$$

Weight fraction (W_i/W_{i-1}) – Cruise

$$\left(\frac{L}{D}\right)_{\text{cruise}} = \frac{C_L}{C_D} = \frac{C_L}{C_{D_0} + C_{Di} + C_{D_{\text{Lift-viscous}}} + C_{D_{wave}}}$$

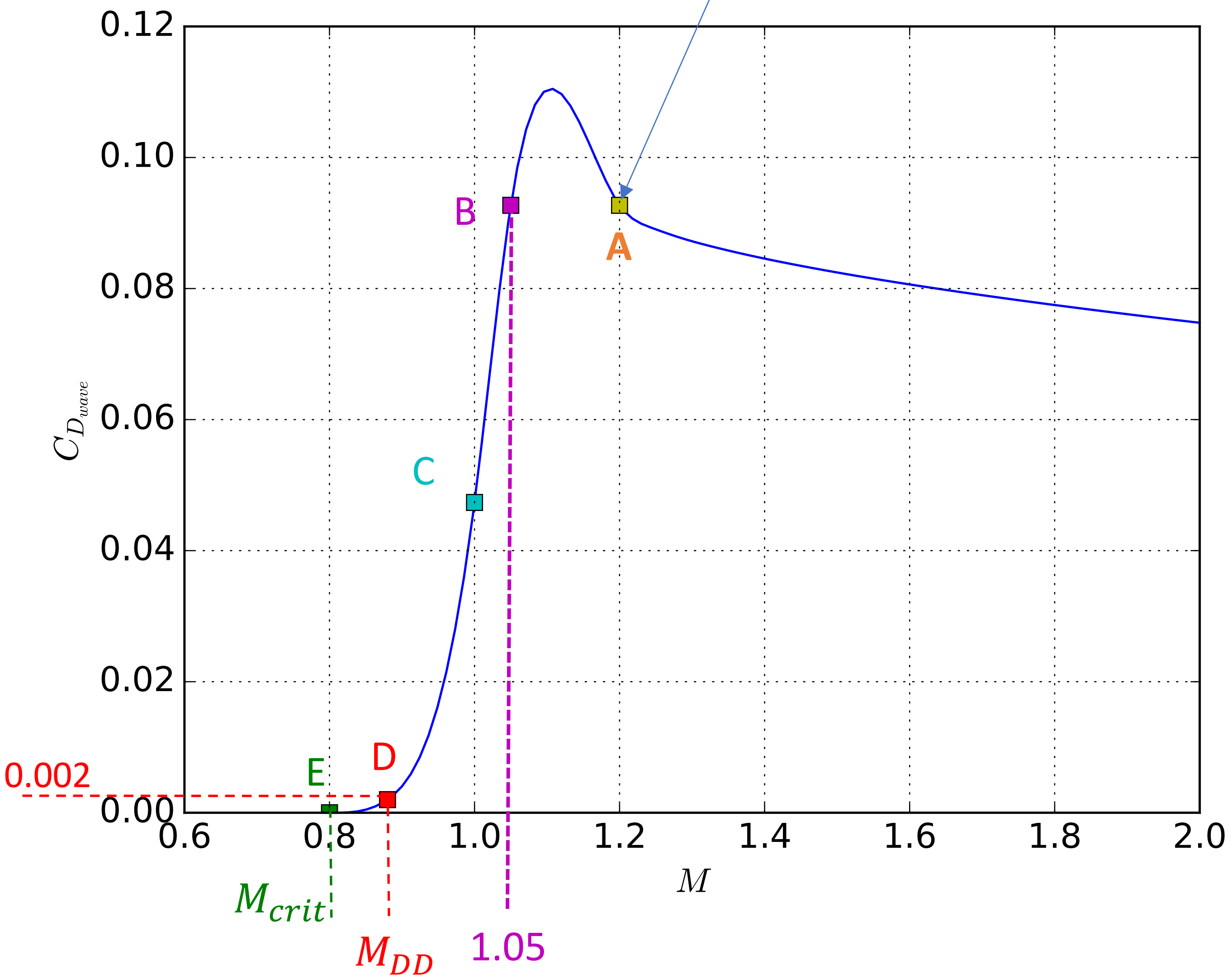
Where:

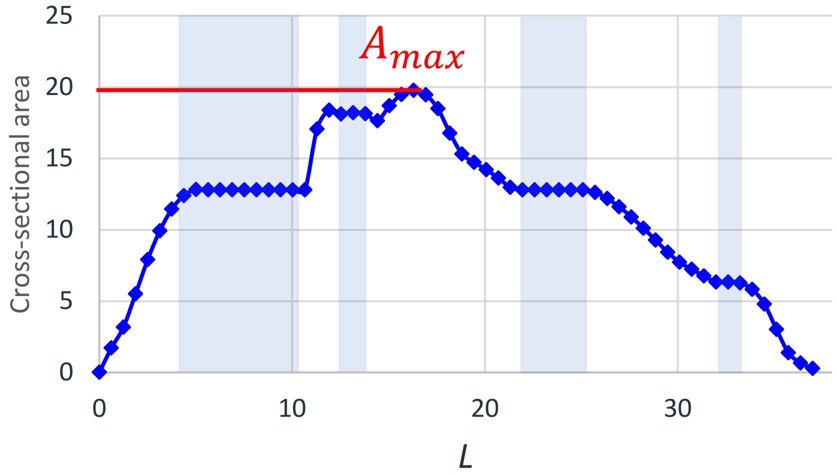
- C_{D0}: Parasite drag, typically 0.015 for transport aircraft (use VSP)
- $C_{D_i} = \frac{1}{\pi A e} C_L^2$, where A is the aspect ratio and e is the Oswald efficiency
- $C_{D_{\text{Lift-viscous}}} = K_0 C_{D0} C_L^2$, where $K_0 = 0.38$ is a factor determined using flight data.
- $C_{D_{wave}}$, is the drag due to compressibility effects usually determined from a a wave drag plot

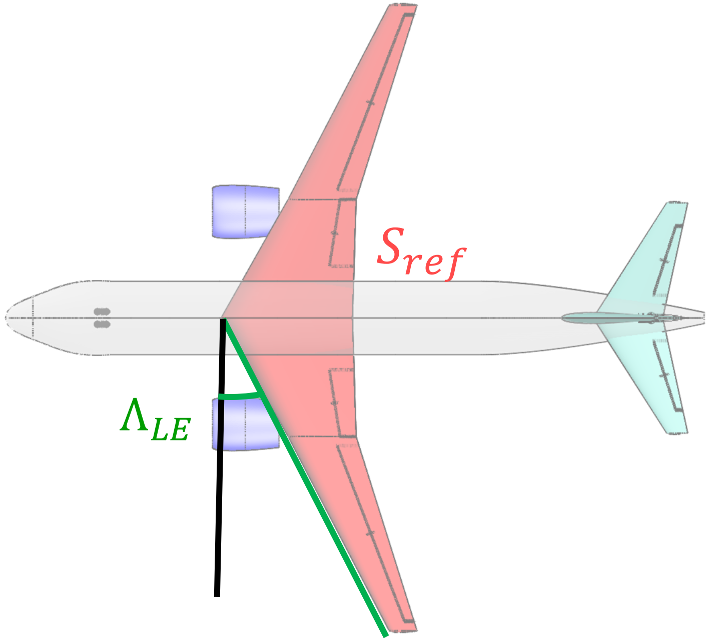
Wave Drag – Proposed method (example: my A320 model)

$$\left(D/q\right)_{wave} = E_{WD}\left[1 - 0.386(M-1.2)^{0.57}\left(1 - \frac{\pi\,\Lambda_{LE-deg}^{0.77}}{100}\right)\right]\left(D/q\right)_{Sears-Haack}$$

$$\left(D/q\right)_{Sears-Haack} = \frac{9\pi}{2}\left(\frac{A_{max}}{l}\right)^2 \qquad l = L - shaded$$







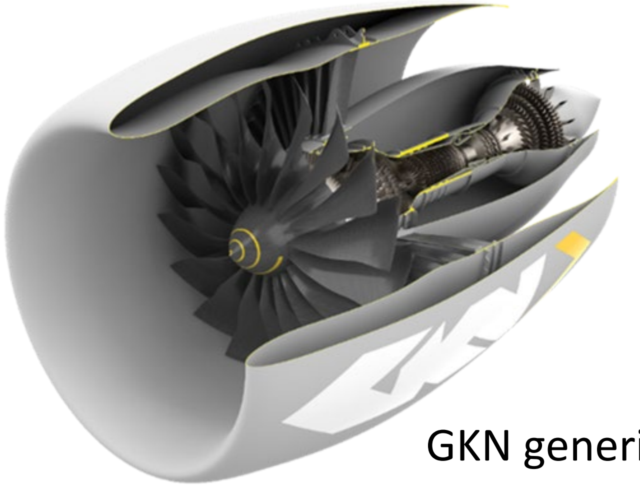
Initial sizing – Engine (rubber) and Wing

Engine:

- $T = \frac{W_0}{N} \cdot g_0 \left(\frac{T}{W}\right)_{\text{takeoff}}$ (Newton)

- $P = \frac{W_0}{N} \left(\frac{P}{W}\right)_{\text{takeoff}}$ (kW, P/W given in kW/kg)

N – number of engines

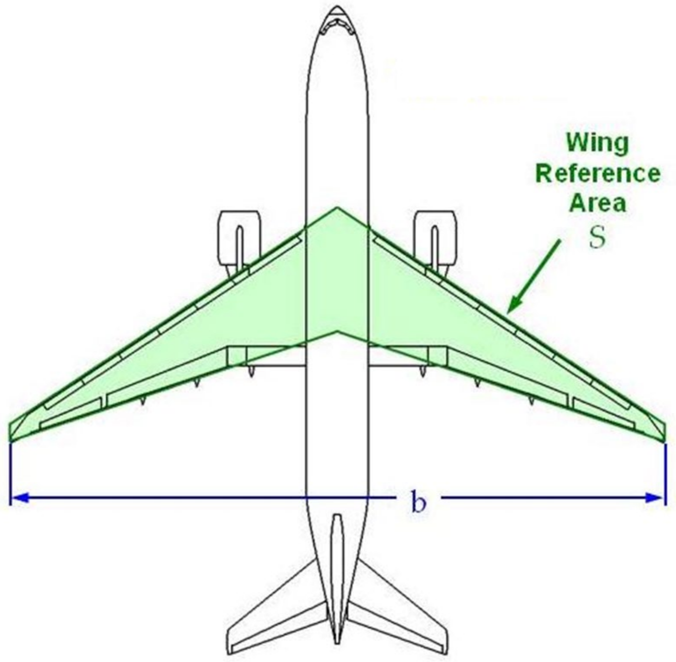


GKN generic turbofan

Wing:

- $S = \frac{W_0}{\left(\frac{W_0}{S}\right)_{\text{takeoff}}}$ (reference area)

- Use eqs. provided in lecture 2 to layout the wing.



Initial sizing – Fixed Engine size

- Existing engines are very often re-used.
- T/W is no longer constant.
- Achieve either mission range or performance – not both

V2500 engine on a 2015 Embraer KC390

V2500 engine on a 1992 A320ceo

Initial sizing – Fixed Engine size

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Constant performance

- Determine T/W – constant
- Calculate $W_0 = \frac{T \cdot N}{\left(\frac{T}{W}\right)_{\mathrm{takeoff}}}$
- Iterate on sizing mission range until W_0 is satisfied

Constant range

- T/W will change as W_0 is computed to meet the required range
- The performance needs to be calculated after establishing T/W

Conceptual Layout

What defines a suitable configuration?

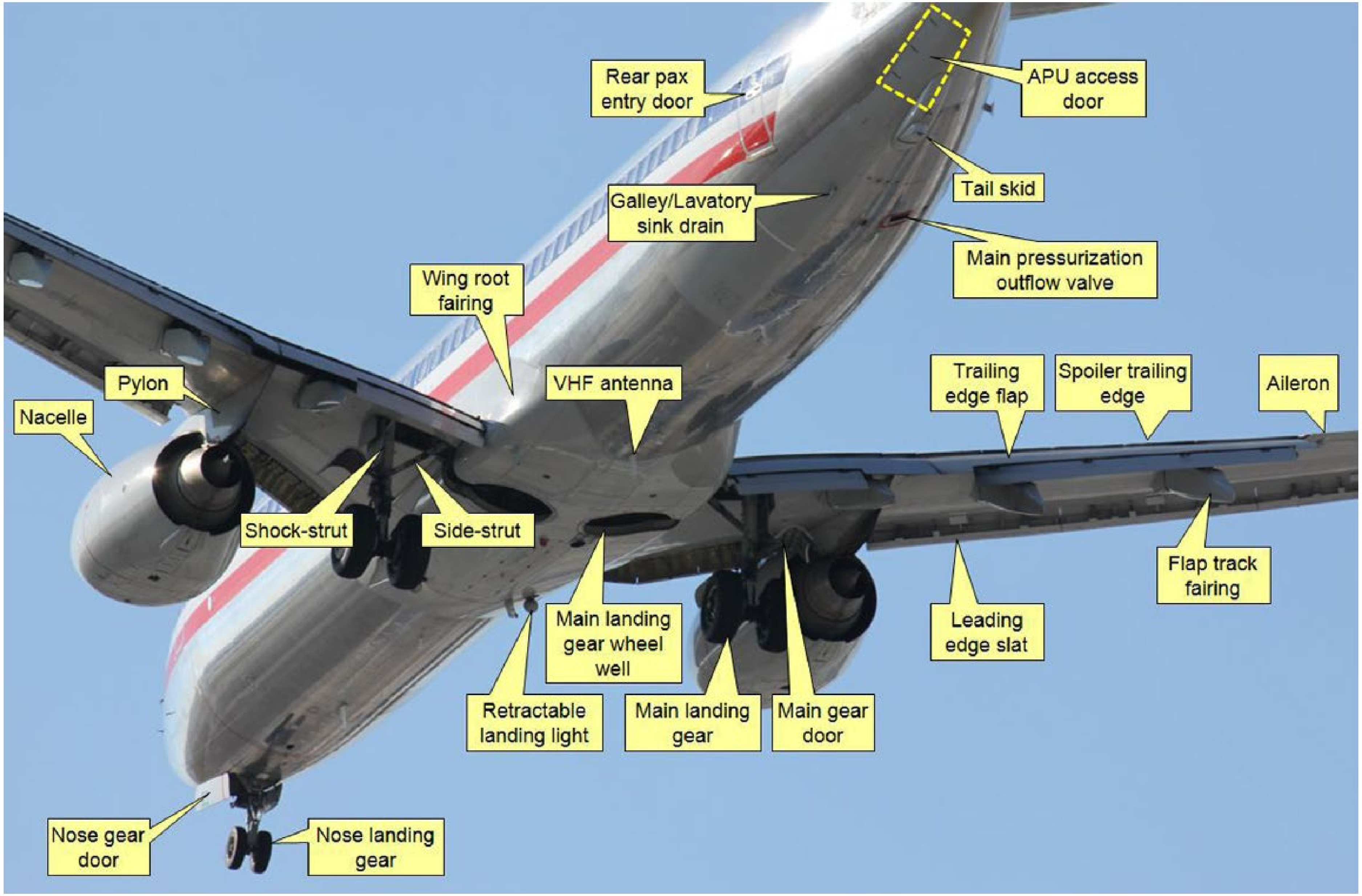
- The requirements, mission definition, and regulations lead the way. => How far? How fast? How high? How heavy? What is the airplane suppose to do? Which regulations apply?

- Past configurations with similar requirements lead to safe development paths. => Proven safety and reliability.
  - https://www.airbus.com/en/airport-operations-and-technical-data/aircraft-characteristics
  - https://www.boeing.com/commercial/airports/plan_manuals.page
  - Q400

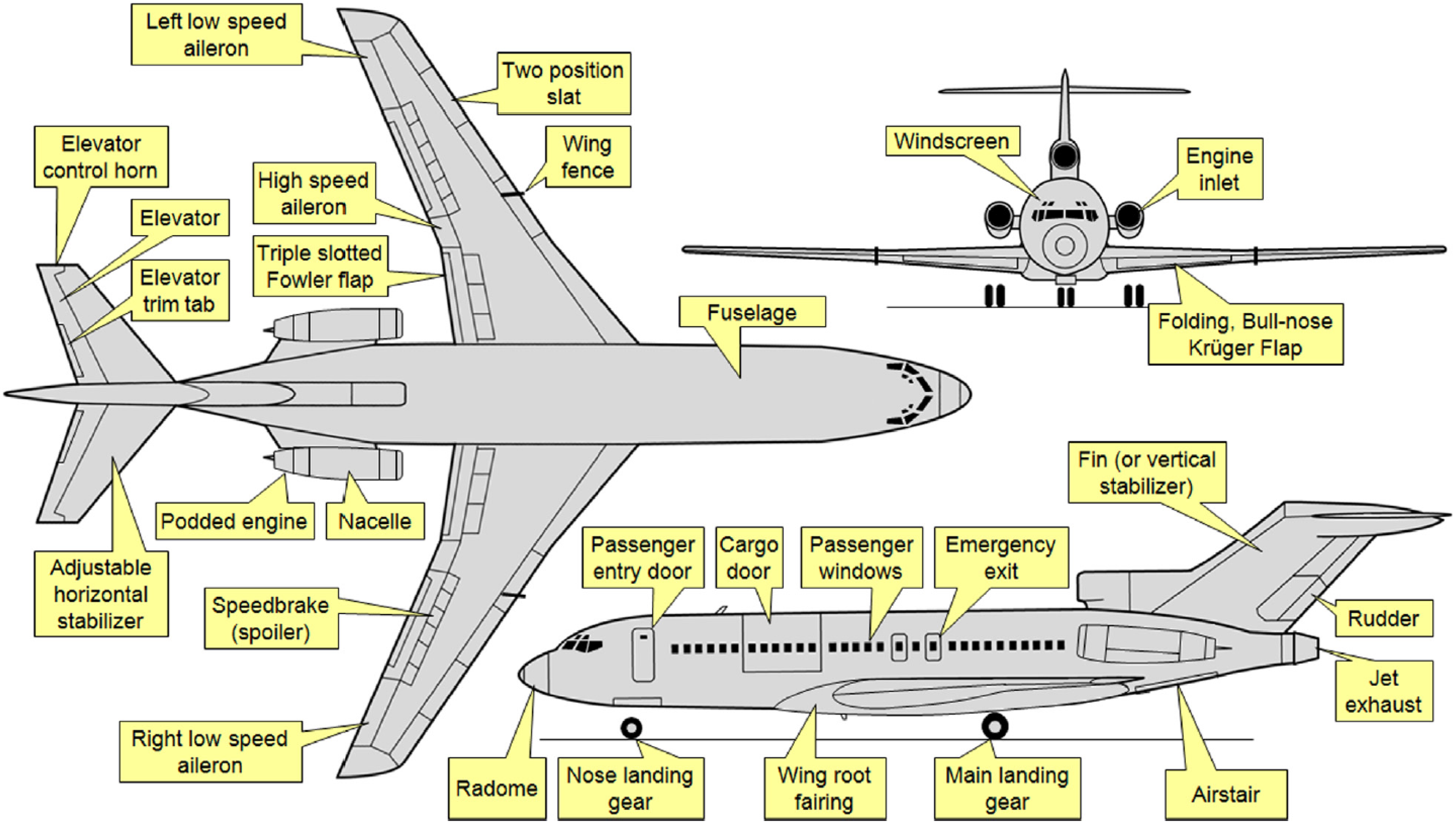
- More radical solutions might lead for better performance, but are risky.

- Nice looking aircraft fly better! (matter of taste).

Aircraft Components – Boeing 737



Aircraft components



Make sure everything fits!

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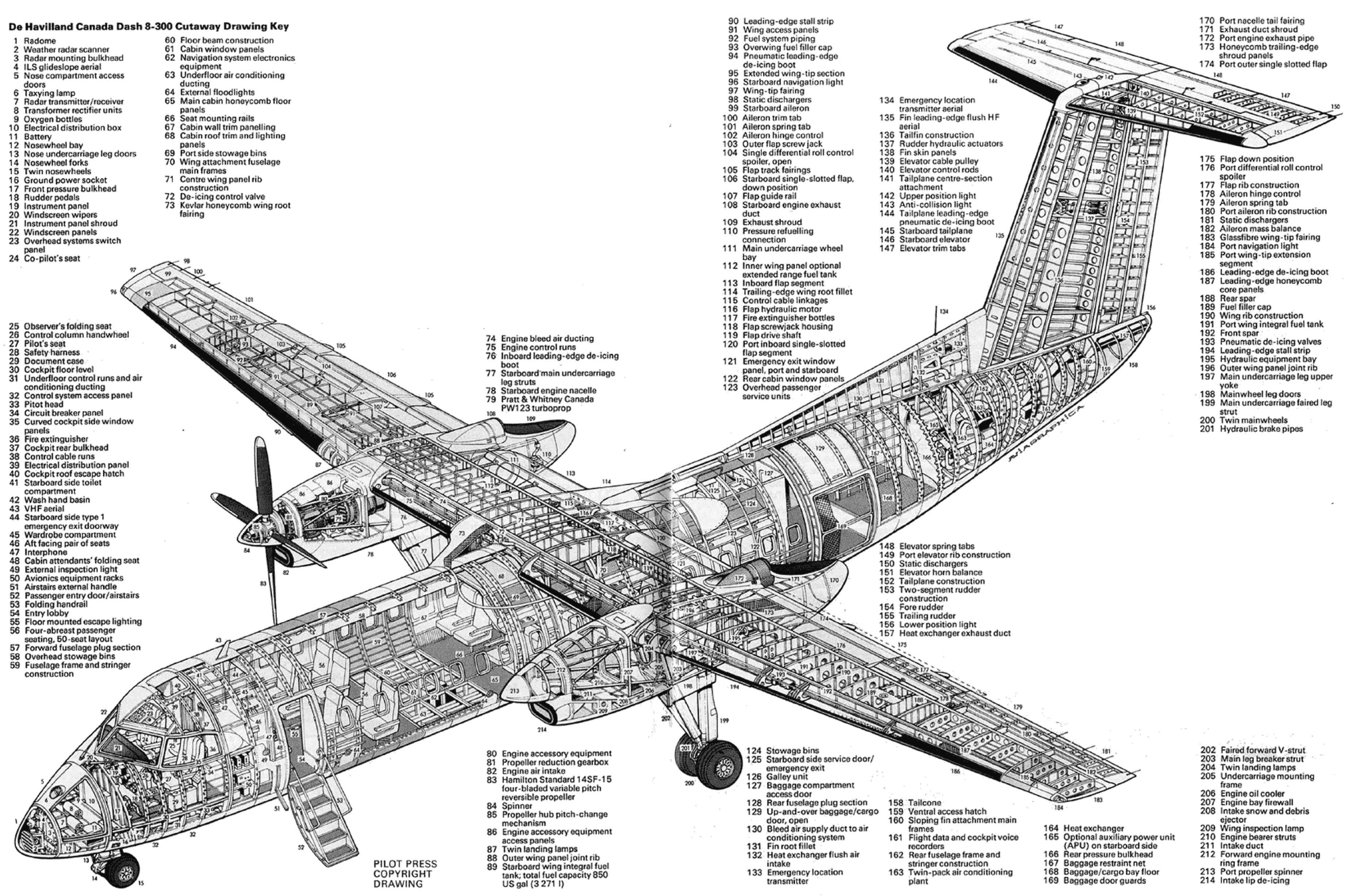
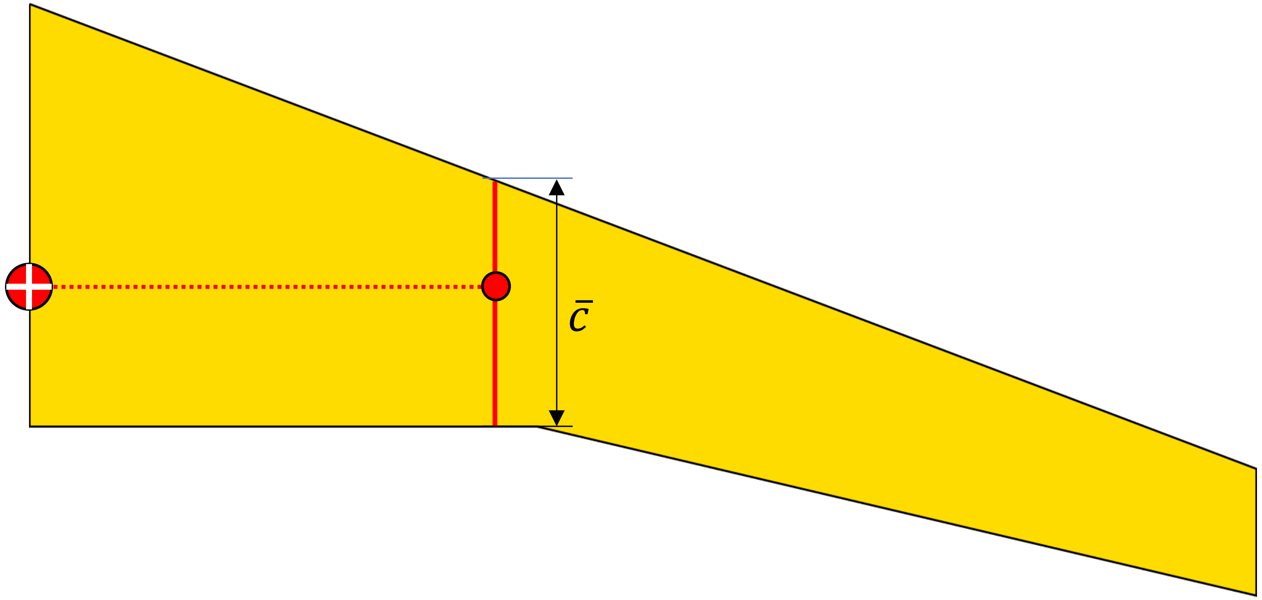
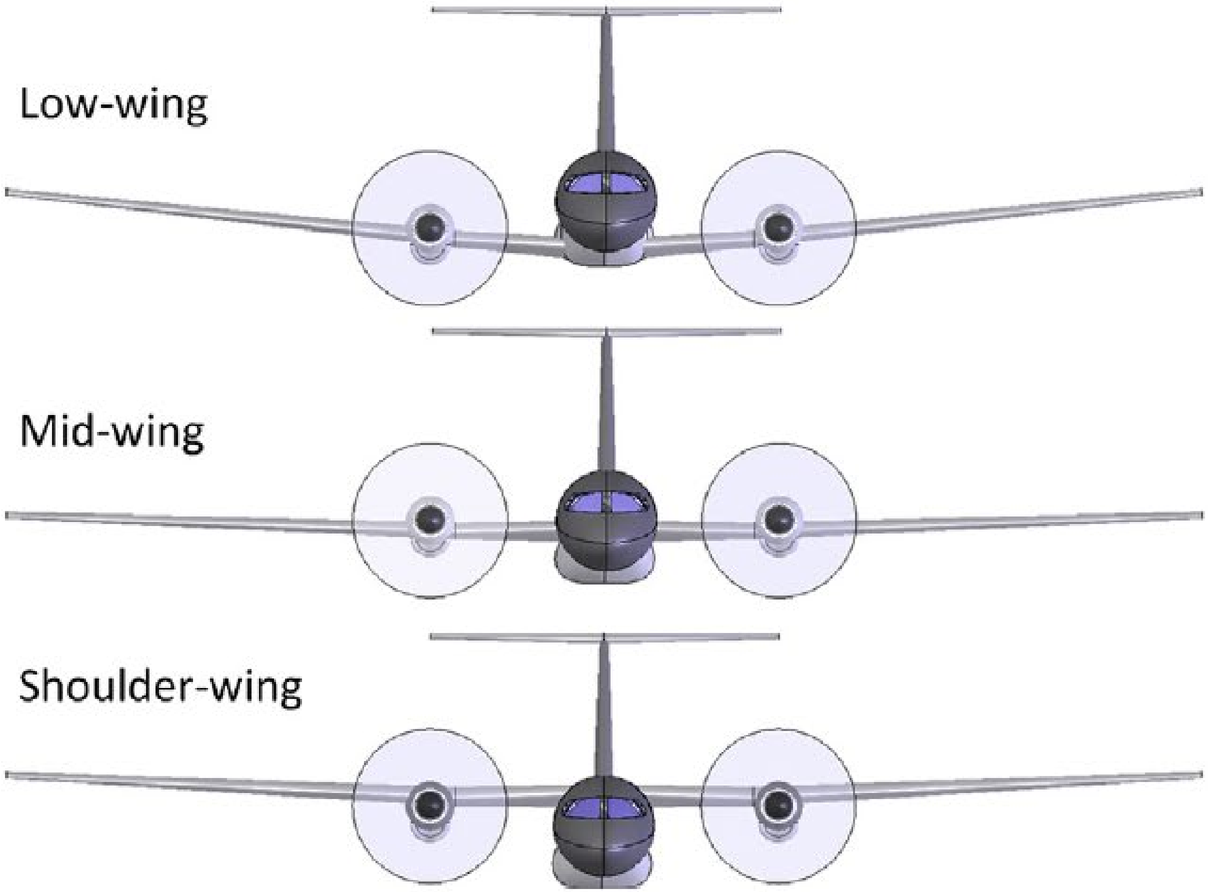


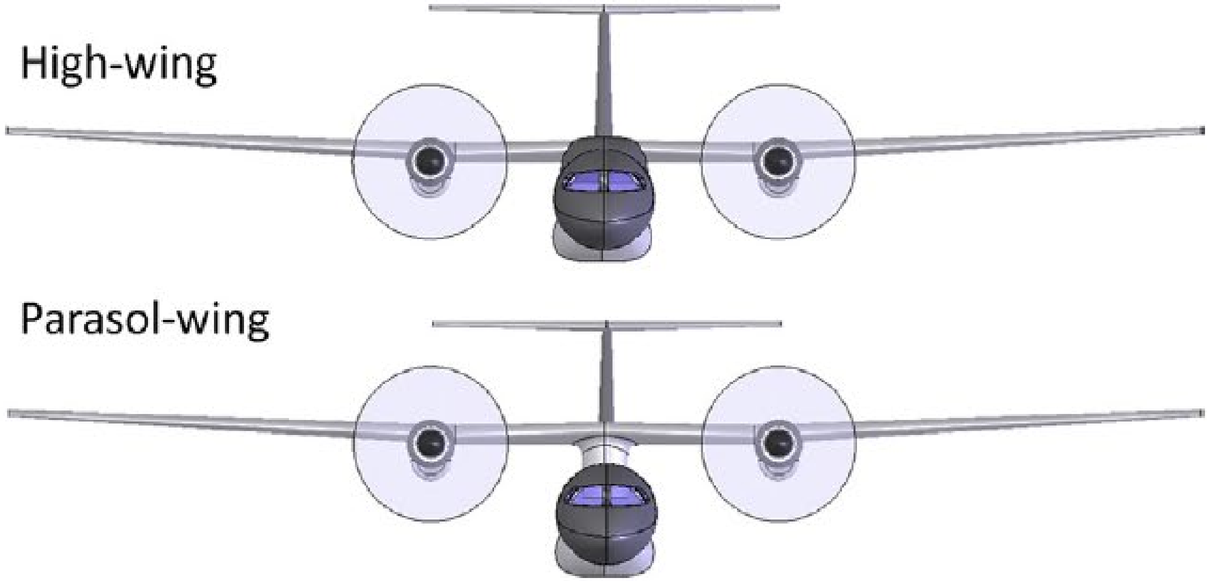


- Flying wing, no tail
  - CG at 25% of $\bar{c}$ (MAC)
- Stable aircraft, with aft-tail:
  - CG at 30% of $\bar{c}$ (MAC)
- Unstable aircraft, with aft-tail:
  - CG and 40% of $\bar{c}$ (MAC)

- Aircraft CG:
  - Estimated after detailed component weight break down.
  - Moves during the mission and for payload variations, **cannot go beyond limits set in CG envelope.**
  - Historically:
    - For front engine aircraft: 35% of fuselage length
    - Engine on wing: 40 – 50% of fuselage length
    - Aft engine: 45 – 55% of fuselage length

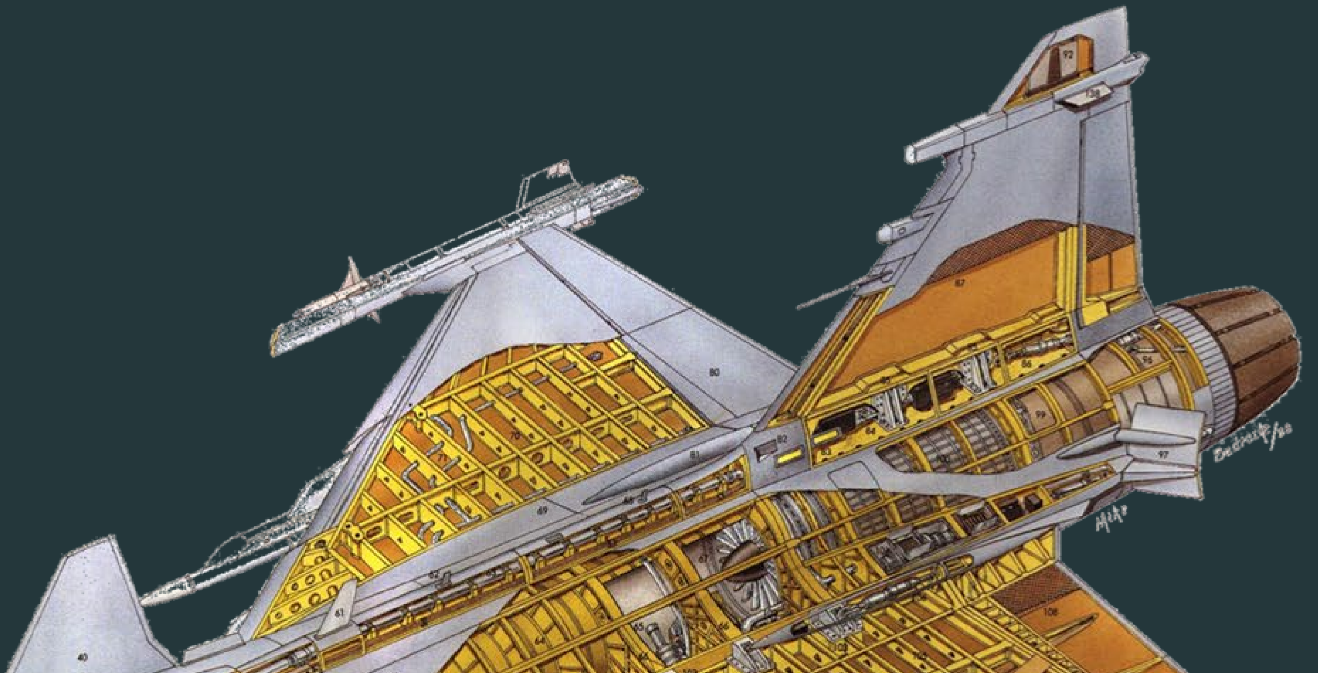
Wing location – Vertical location





Source: Snorri G (2014), *General aviation aircraft design: Applied methods and procedures*, Butterworth-Heinemann, 1st Ed.

Aerodynamic and structural considerations in layout

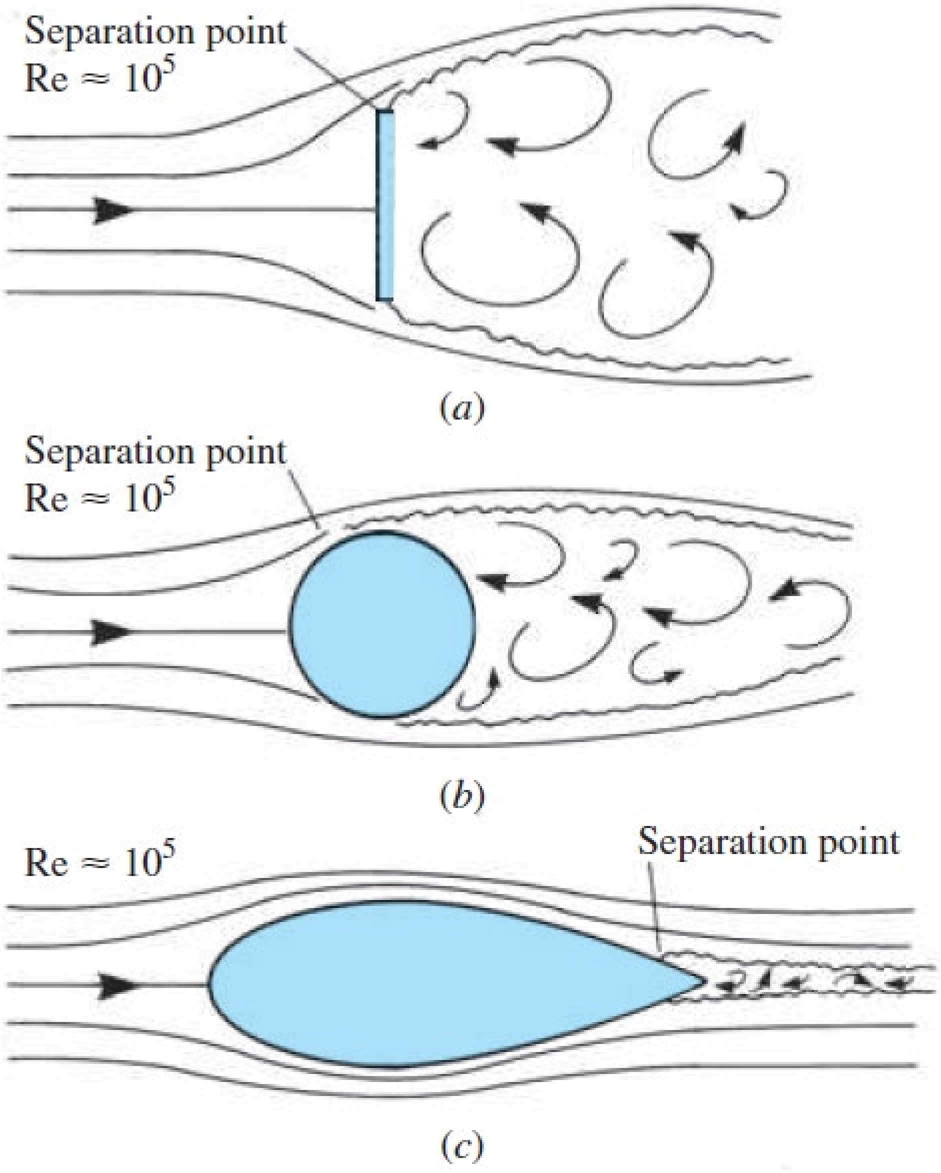


To improve aerodynamics:

- Reduce frontal and **wetted area**.
- Clean configuration with minimum interference drag.
- Adequate fineness ratio – (6-8 for subsonic aircraft):

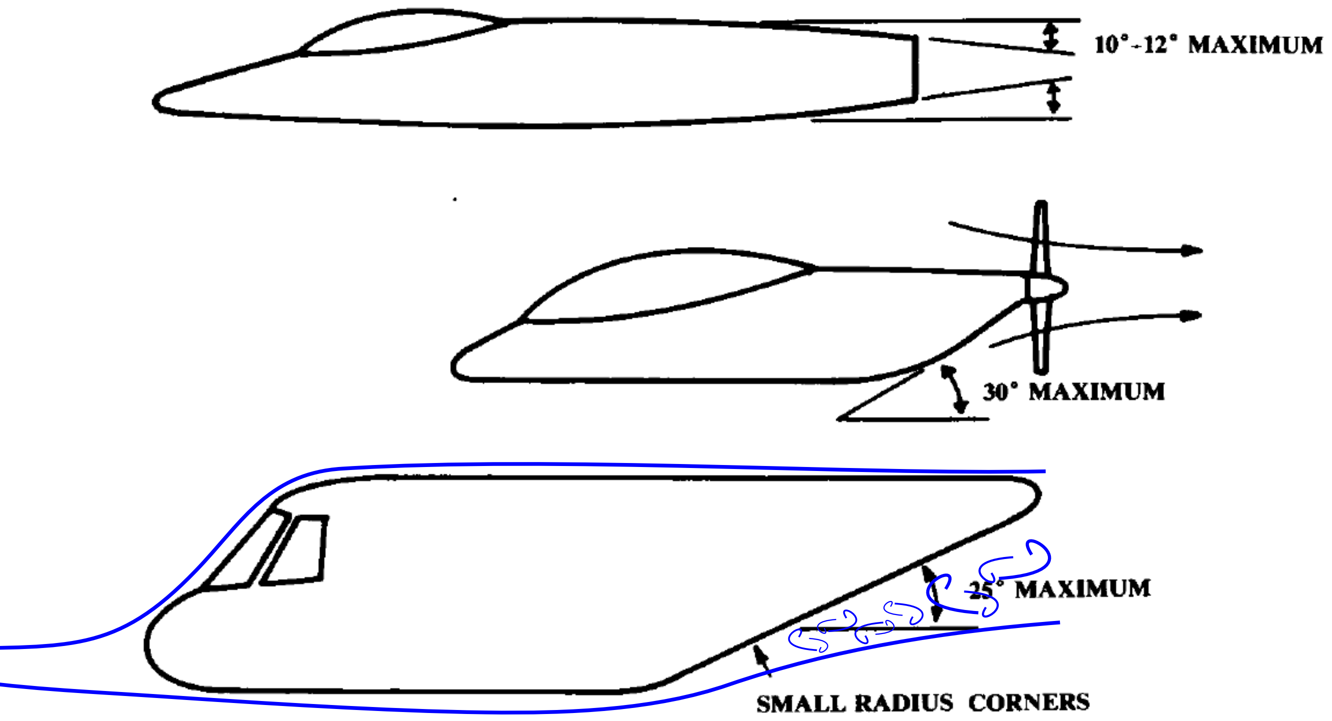
$$f = \frac{l_{fuse}}{d_{\max}}$$

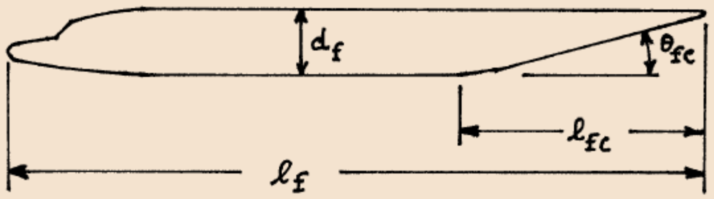
- Avoid sharp corners.
- An experienced designer can ”see air”.



From Andersson, Fundamentals of Aerodynamics

Shaping the fuselage:





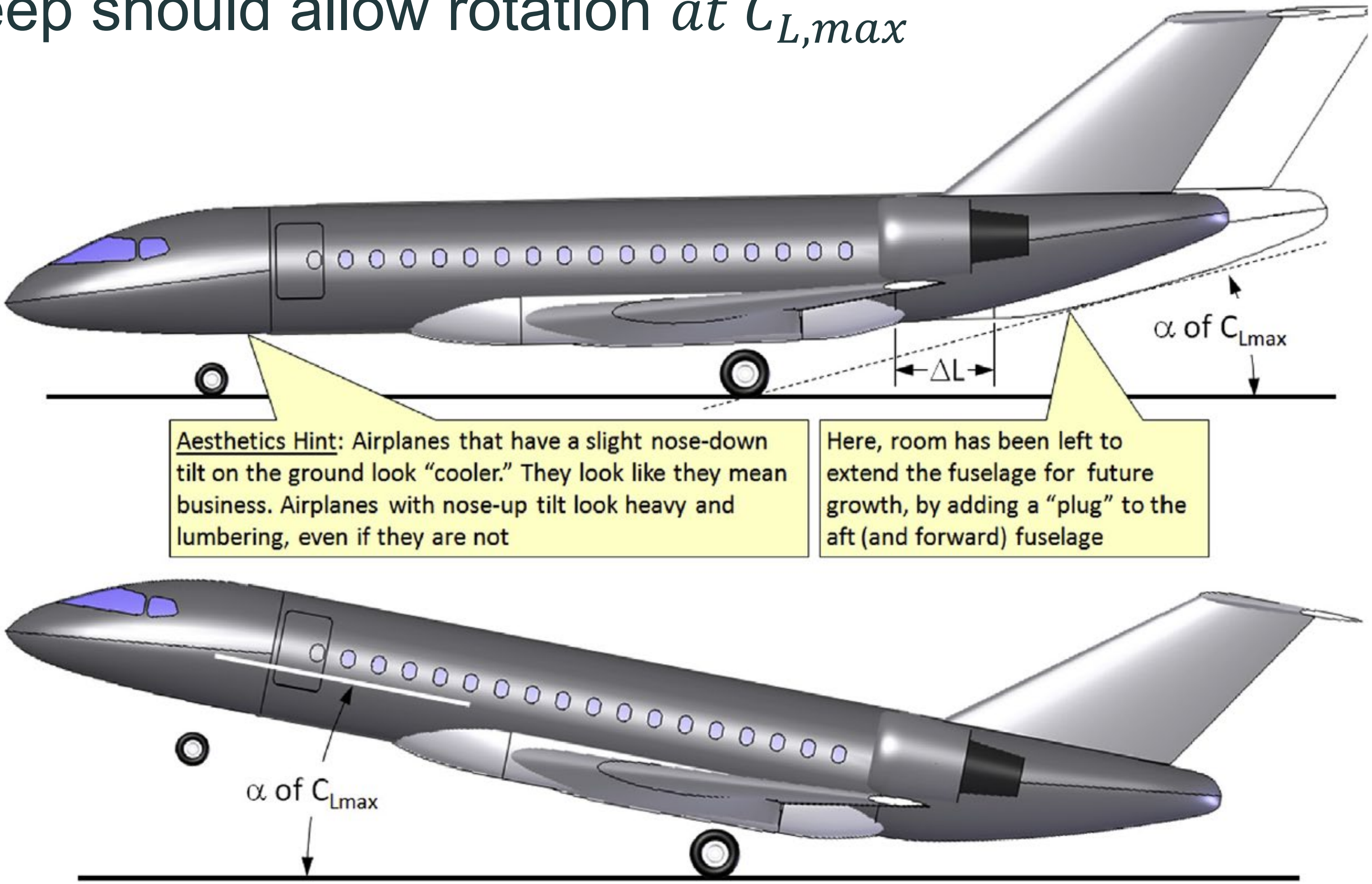
Definition of Geometric Fuselage Parameters

Currently Used Geometric Fuselage Parameters

Airplane Type	l_f/d_f	l_{fc}/d_f	θ_{fc} (deg)
Homebuilts	4 - 8	3	2 - 9
Single Engine	5 - 8	3 - 4	3 - 9
Twins	3.6 - 8	2.6 - 4	6 - 13
Agricultural	5 - 8	3 - 4	1 - 7
Business Jets	7 - 9.5	2.5 - 5	6 - 11
Regionals	5.6 - 10	2 - 4	15 - 19
Jet Transports	6.8 - 11.5	2.6 - 4	11 - 16
Mil. Trainers	5.4 - 7.5	3	up to 14
Fighters	7 - 11	3 - 5	0 - 8
Mil. Transports, Bombers and Patrol Airplanes	6 - 13	2.5 - 6	7 - 25
Flying Boats	6 - 11	3 - 6	8 - 14
Supersonics	12 - 25	6 - 8	2 - 9

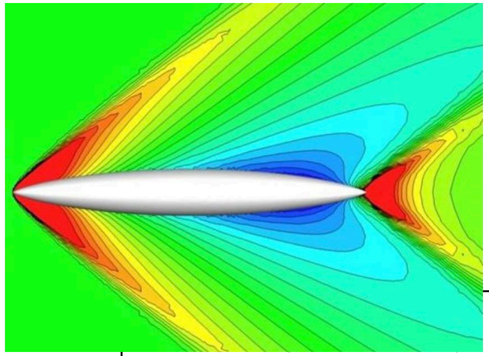
From Roskam, Airplane Design: Part 2 - preliminary configuration design and integration of the propulsion system

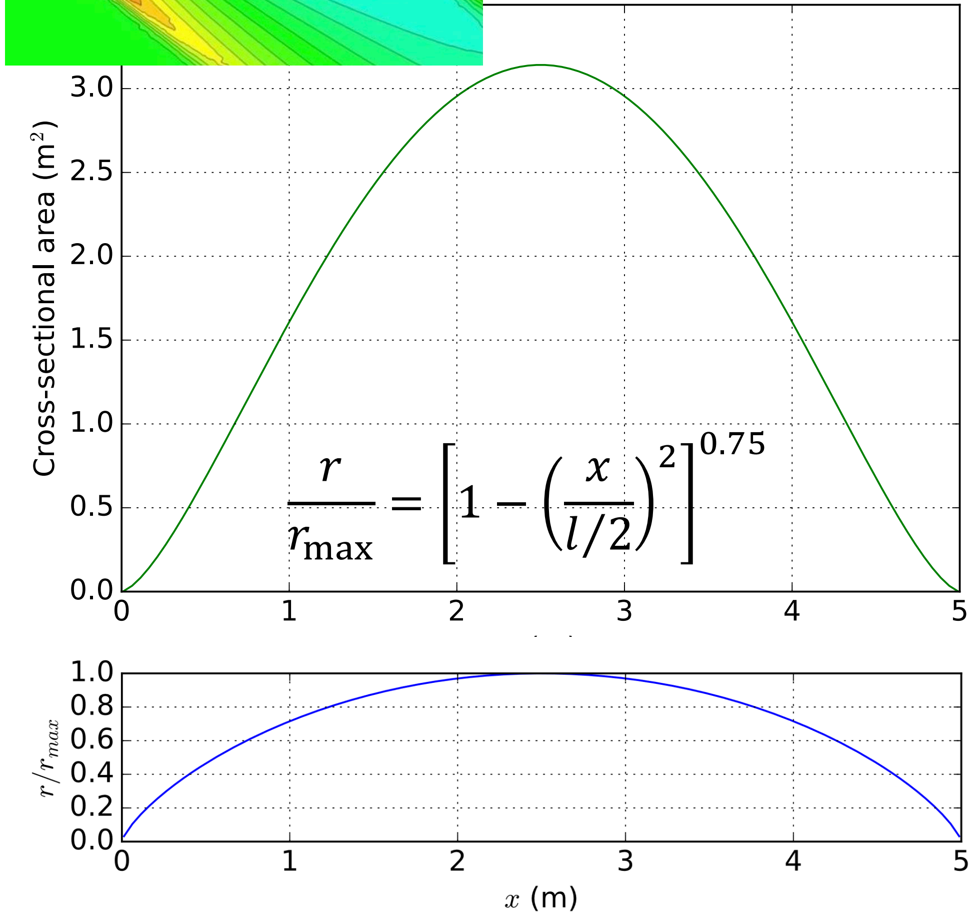
The upsweep should allow rotation $at\ C_{L,max}$



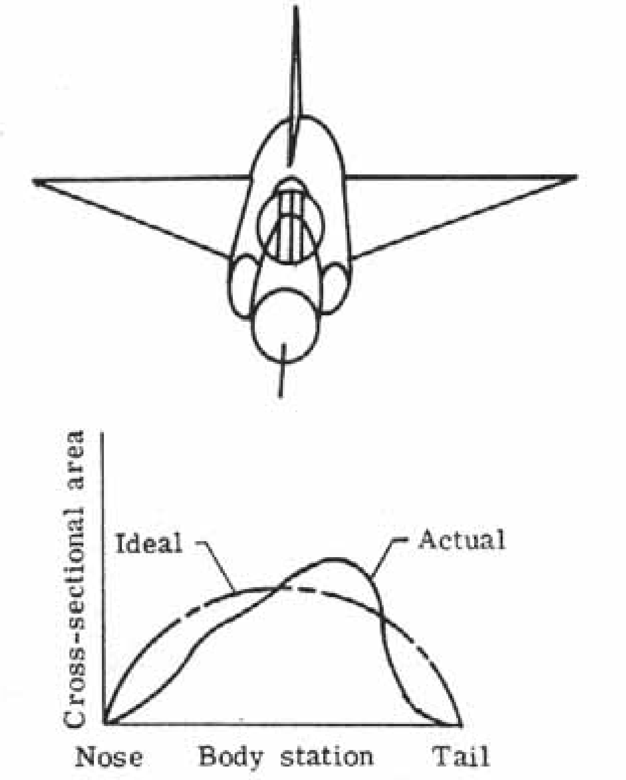
Aerodynamic considerations – longitudinal for wave drag

- Wave drag: caused by pressure drag due to shock waves.
- Wave drag increases with curvature, 2nd derivate of volume plot.
- The Sears-Hack body gives the lowest wave drag for M=1.
- Volume distribution in aircraft cannot follow that shape => where do we put the wings!

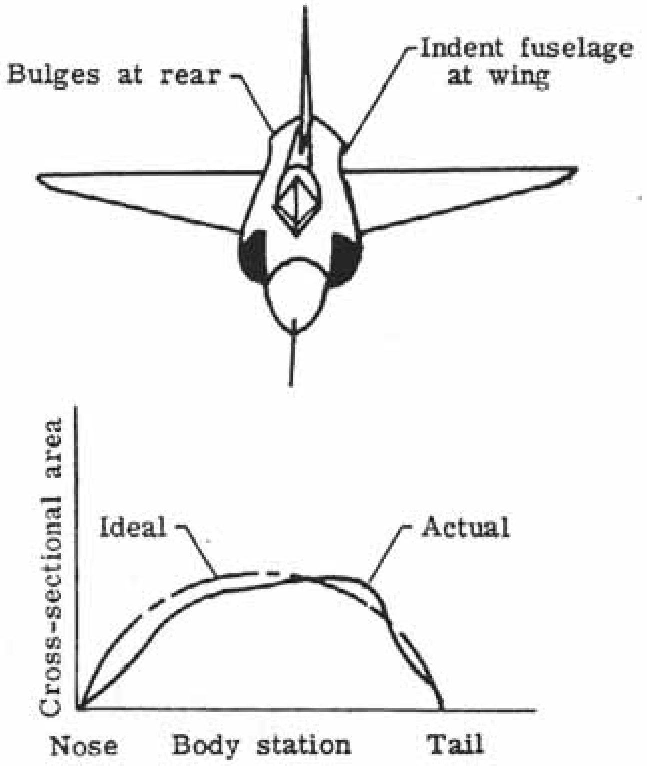




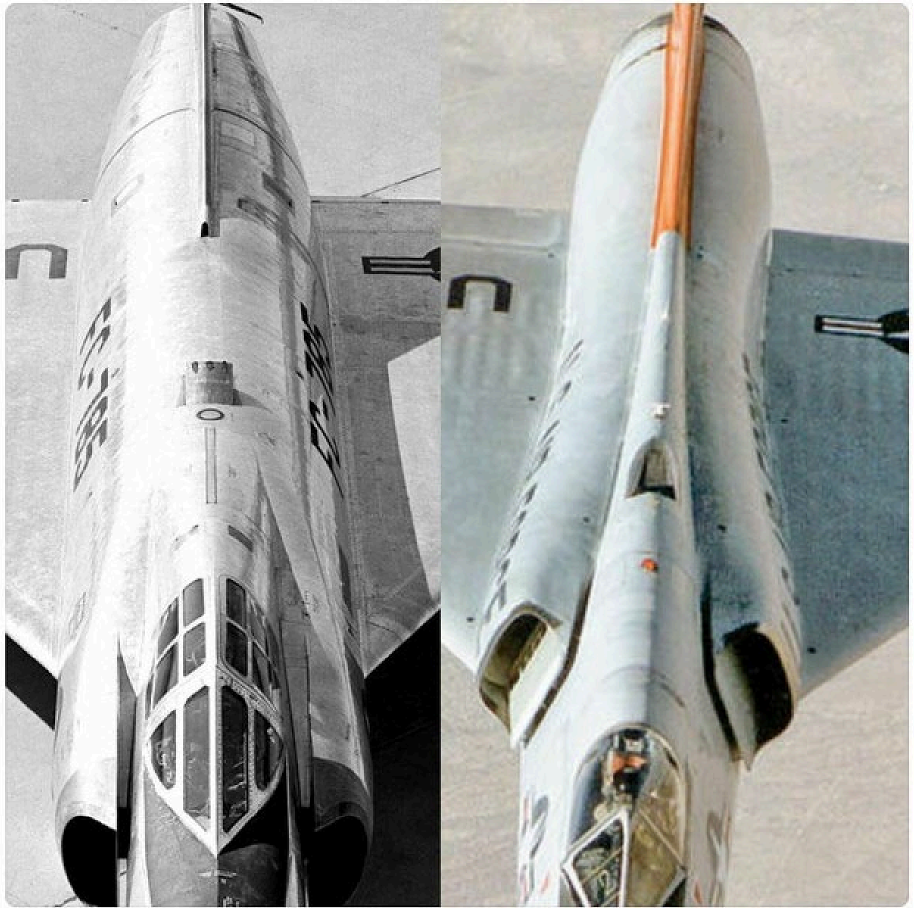
Supersonic area rule, reductions in wave drag up to 50%



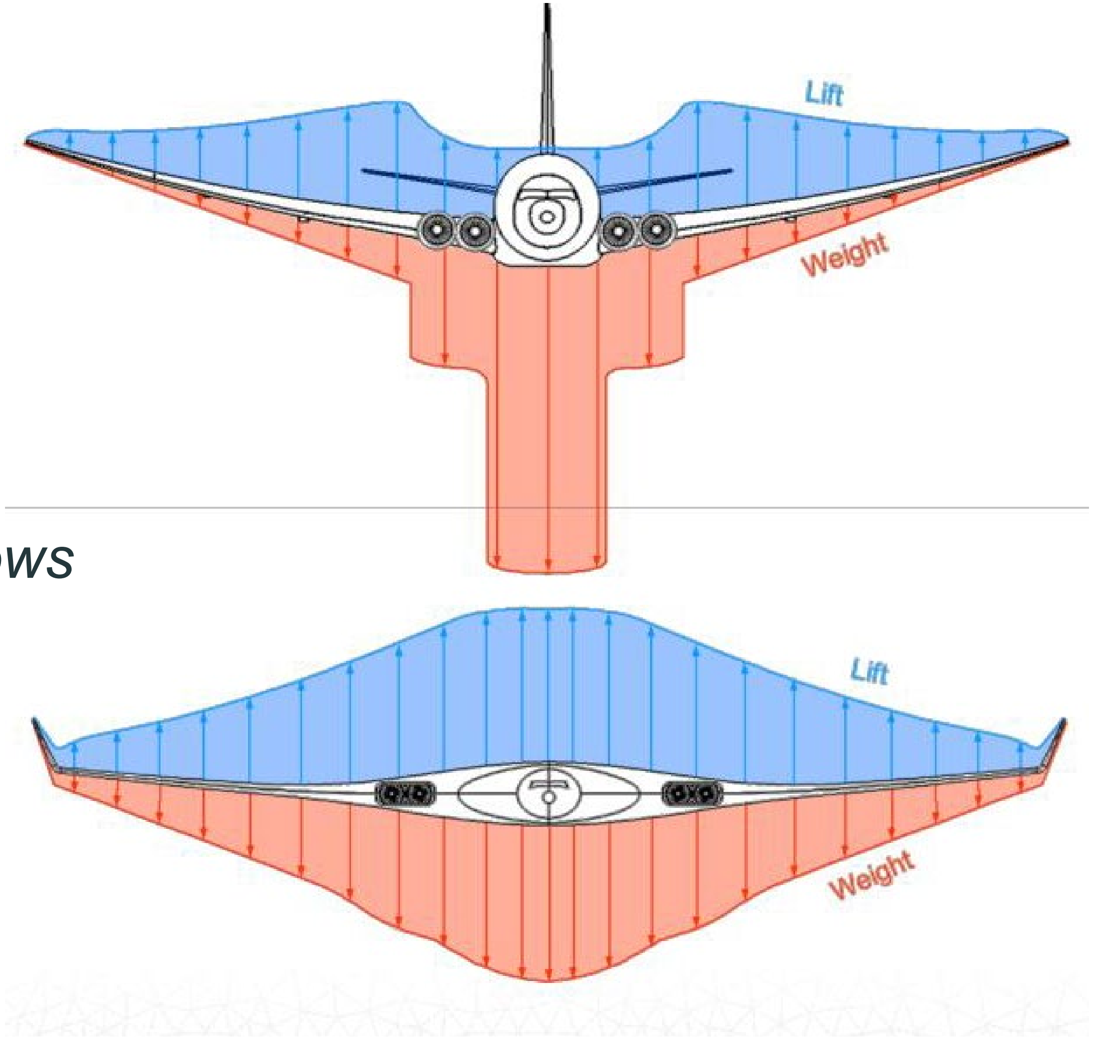
(a) YF-102A before area ruling.



(b) F-102A after area ruling.



- Short, straight and continuous load paths.
  - Major loads on few major components
  - Heavy items near CG
- Minimum cutouts and doors.
  - Avoid engines buried in fuselage
  - Don't place the landing gear in the wingbox!
  - Beware of stress concentrations. *Why are windows rounded?*
- Avoid T-tails, H-tails and all moving tails.
- Span loading for weight reduction.
  - Distribute your loads in a smart way!



Structural considerations – Carrythrough structure

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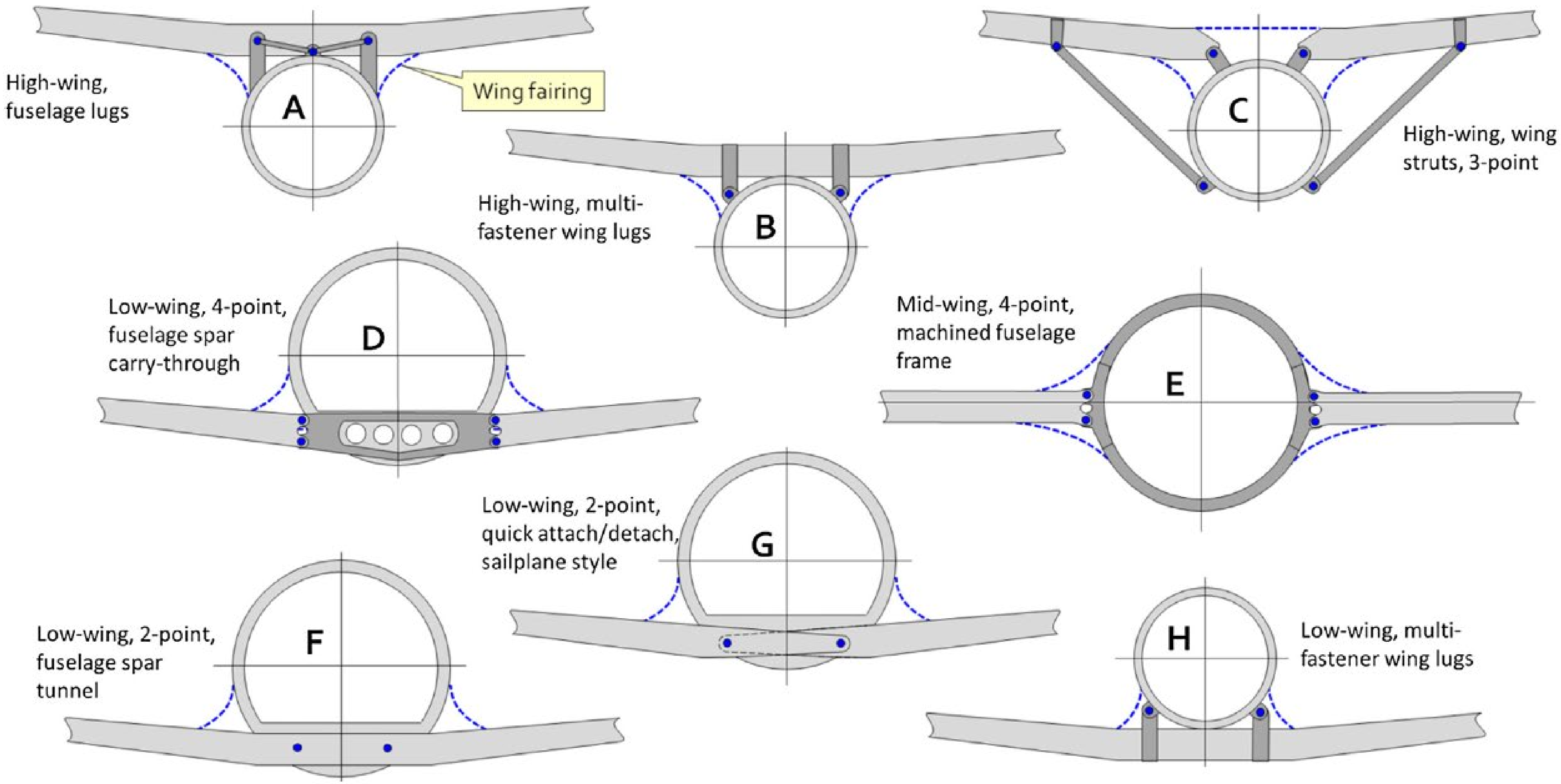

Other considerations – Crashworthiness

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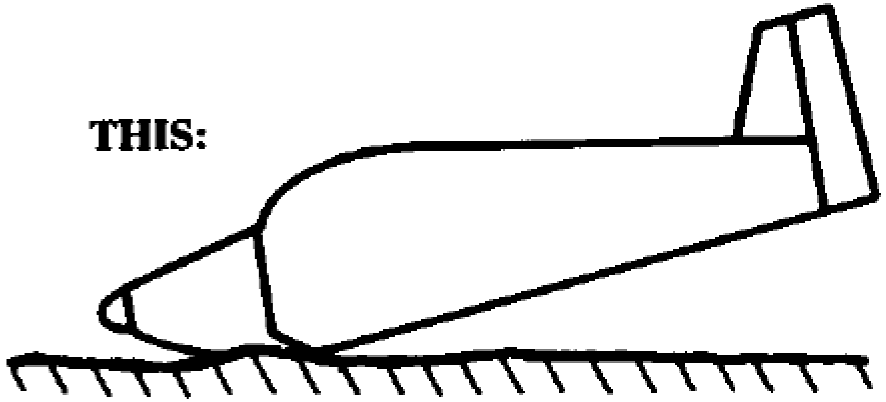

SCARFED FIREWALL PREVENTS SCOOPING

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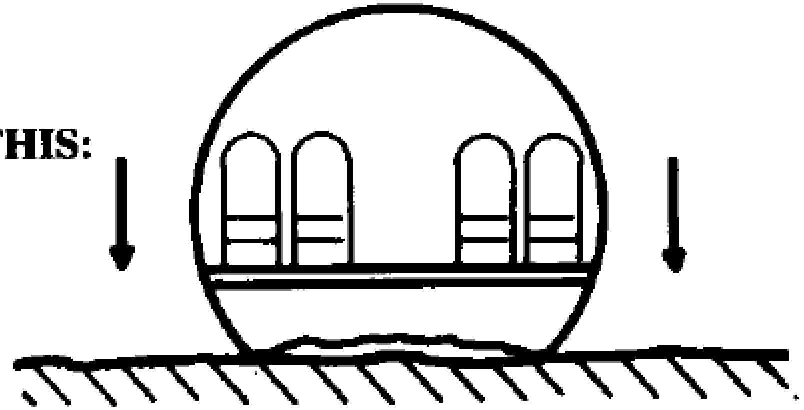
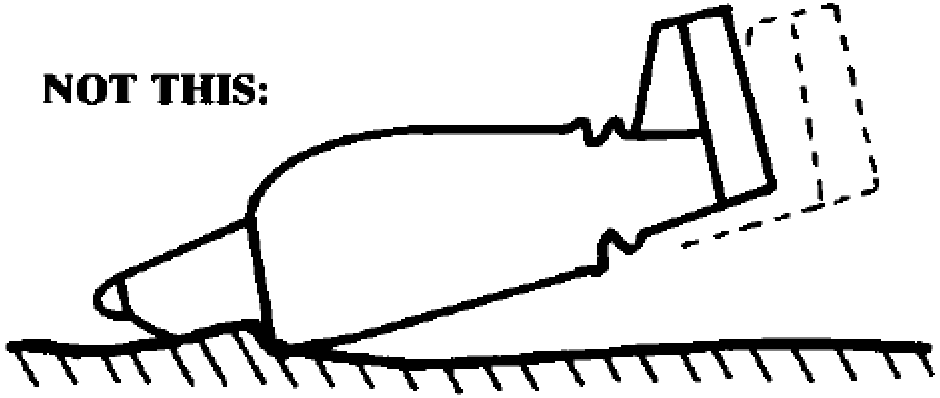


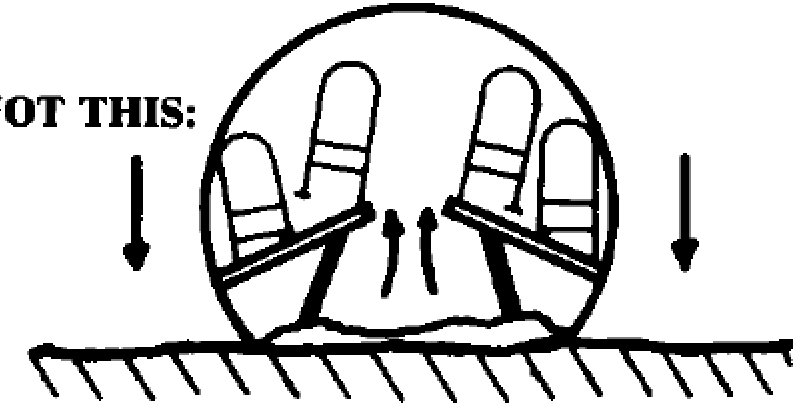
NO FLOOR STRUTS

NOT THIS:

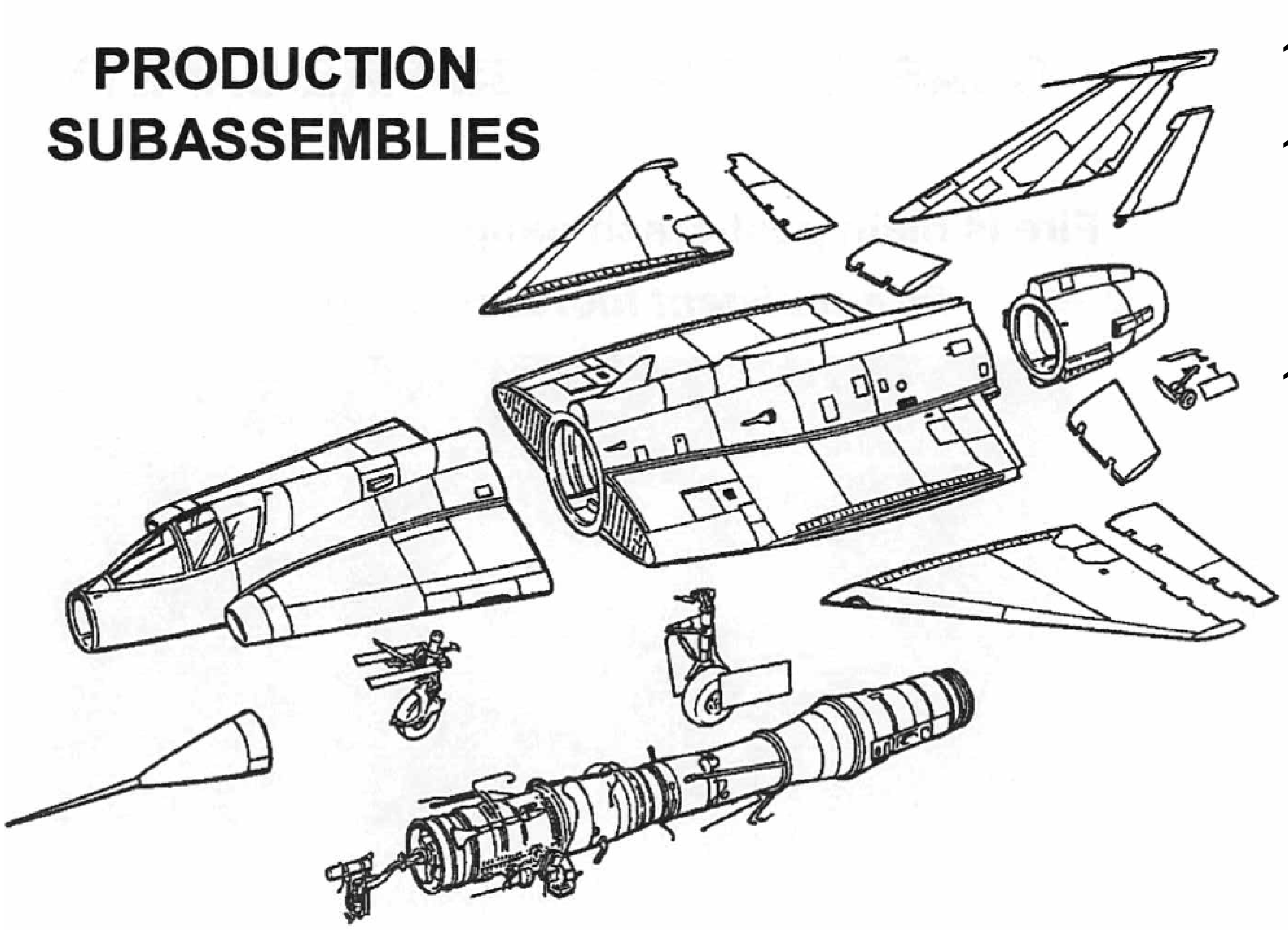


FIREWALL SCOOPING INCREASES CRASH LOADS

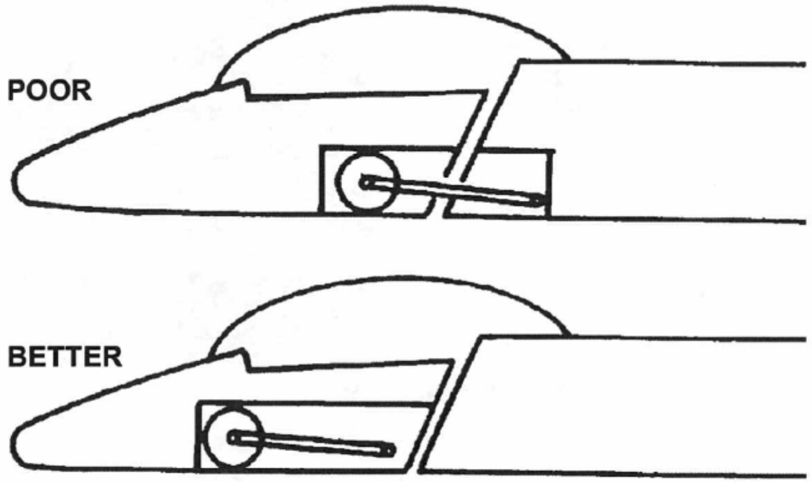
NOT THIS:



FLOOR STRUTS PUSH
FLOOR UPWARDS



- Reduced part count.
- Accessibility
  - Mechanics need to stick their head in tight places
- Lower cost of materials and processes



Crew station, cabin and payload

Important dimenssions!

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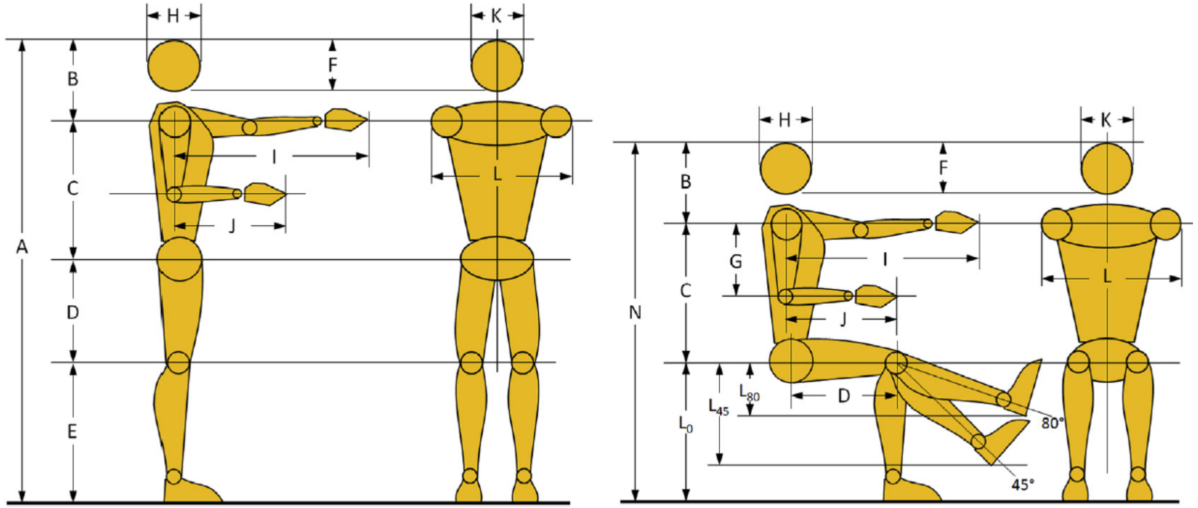
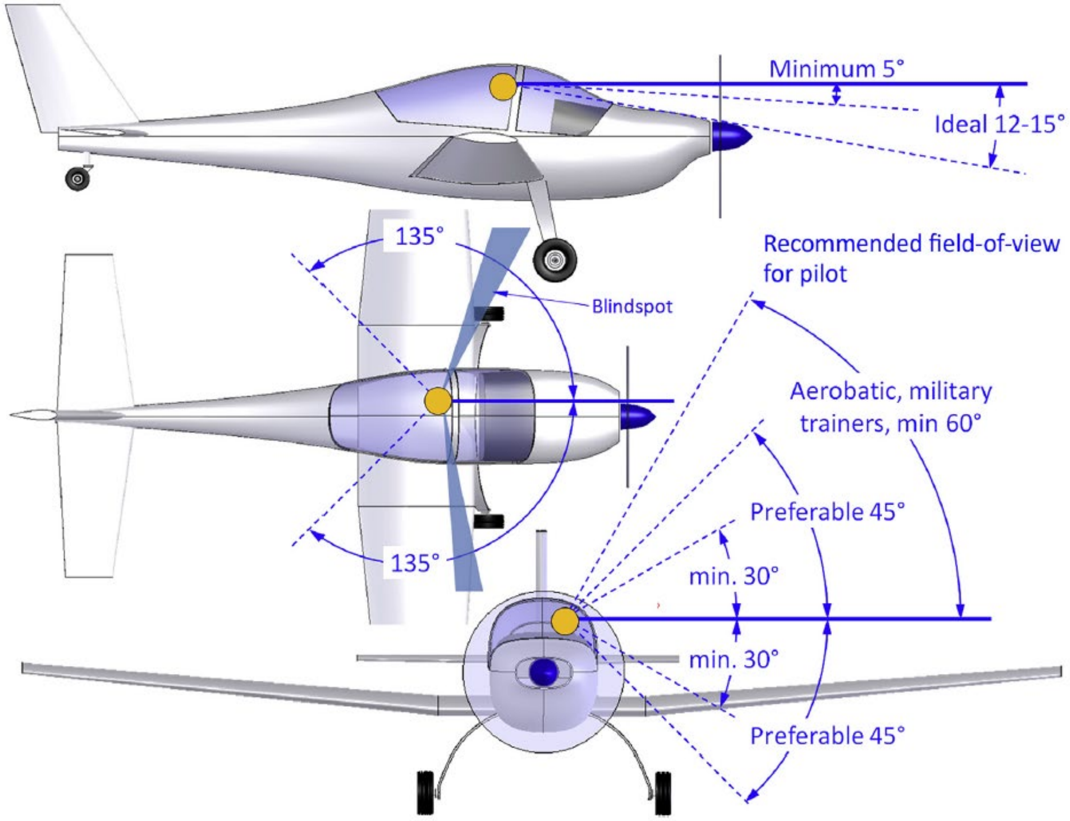
			MALE AND FEMALE PHYSICAL CHARACTERISTICS					
			1st	5th	50th	95th	99th	
Stature	A	Male	63.1	64.8	69.1	73.5	75.2	in
			160.3	164.6	175.5	186.7	191.0	cm
		Female	58.4	60.2	64.1	68.4	70.1	in
			148.3	152.9	162.8	173.7	178.1	cm
Shoulder-head	B	Male	11.8	11.9	12.5	13.0	13.2	in
			30.0	30.2	31.8	33.0	33.5	cm
		Female	10.9	11.3	11.8	12.3	12.5	in
			27.7	28.7	30.0	31.2	31.8	cm
Torso	C	Male	19	19.3	20.1	20.8	21.1	in
			48.3	49.0	51.1	52.8	53.6	cm
		Female	17.5	17.8	18.4	19.2	19.2	in
			44.5	45.2	46.7	48.8	48.8	cm
Hip height	D	Male	14.9	15.4	16.7	18.0	18.5	in
			37.8	39.1	42.4	45.7	47.0	cm
		Female	14.3	14.7	15.9	17.1	17.8	in
			36.3	37.3	40.4	43.4	45.2	cm
Knee height	E	Male	17.4	18.2	19.8	21.7	22.4	in
			44.2	46.2	50.3	55.1	56.9	cm
		Female	15.7	16.4	18.0	19.8	20.6	in
			39.9	41.7	45.7	50.3	52.3	cm
Head height	F	Male	8.4	8.6	8.6	9.1	9.4	in
			21.3	21.8	21.8	23.1	23.9	cm
		Female	7.8	8.3	8.6	9.1	9.4	in
			19.8	21.1	21.8	23.1	23.9	cm
Upper arm	G	Male	11.3	11.5	12.3	13.5	14	in
			28.7	29.2	31.2	34.3	35.6	cm
		Female	10.5	10.7	11.5	12.4	12.8	in
			26.7	27.2	29.2	31.5	32.5	cm
Head length	H	Male	7.1	7.3	7.8	8.2	8.4	in
			18.0	18.5	19.8	20.8	21.3	cm
		Female	6.8	7.0	7.4	7.8	8.0	in
			17.3	17.8	18.8	19.8	20.3	cm
Functional reach	I	Male	28.4	29.1	31.5	34.1	35.3	in
			72.1	73.9	80.0	86.6	89.7	cm
		Female	25.9	26.7	28.9	31.4	32.4	in
			65.8	67.8	73.4	79.8	82.3	cm
Elbow-fingertip	J	Male	17.1	17.6	19.2	20.6	21.3	in
			43.4	44.7	48.8	52.3	54.1	cm
		Female	15.4	16.0	17.4	19.0	19.6	in
			39.1	40.6	44.2	48.3	49.8	cm
Head width	K	Male	5.1	5.6	6.0	6.3	6.5	in
			13.0	14.2	15.2	16.0	16.5	cm
		Female	5.2	5.4	5.7	6.0	6.1	in
			13.2	13.7	14.5	15.2	15.5	cm
Shoulder breadth	L	Male	17.1	17.7	19.3	21.1	21.7	in
			43.4	45.0	49.0	53.6	55.1	cm
		Female	15	15.6	17.0	18.6	19.4	in
			38.1	39.6	43.2	47.2	49.3	cm



			MALE AND FEMALE PHYSICAL CHARACTERISTICS					
			1st	5th	50th	95th	99th	
Sitting stature	N	Male	48.2	49.4	52.4	55.5	56.7	in
			122.4	125.5	133.1	141.0	144.0	cm
		Female	44.1	45.5	48.2	51.3	52.3	in
			112.0	115.6	122.4	130.3	132.8	cm
Leg at 0°	L_0	Male	17.4	18.2	19.8	21.7	22.4	in
			44.2	46.2	50.3	55.1	56.9	cm
		Female	15.7	16.4	18	19.8	20.6	in
			39.9	41.7	45.7	50.3	52.3	cm
Leg at 45°	L_{45}	Male	12.3	12.9	14.0	15.3	15.8	in
			31.3	32.7	35.6	39.0	40.2	cm
		Female	11.1	11.6	12.7	14.0	14.6	in
			28.2	29.5	32.3	35.6	37.0	cm
Leg at 80°	L_{80}	Male	3.0	3.2	3.4	3.8	3.9	in
			7.7	8.0	8.7	9.6	9.9	cm
		Female	2.7	2.8	3.1	3.4	3.6	in
			6.9	7.2	7.9	8.7	9.1	cm

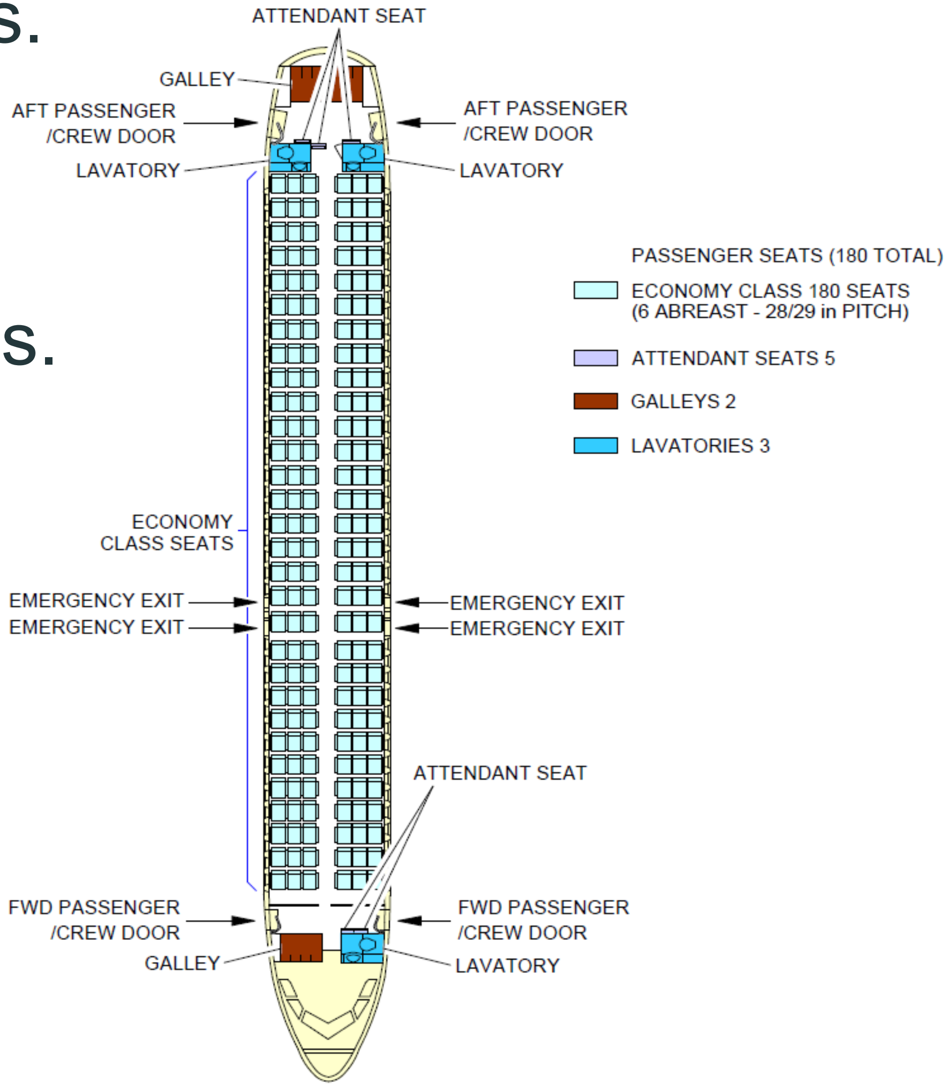
Crew station – important aspects

- Defines the fuselage shape in the vicinity of the cockpit.
- Mainly regulated by vision requirements
- For sizing look into existing aircraft.



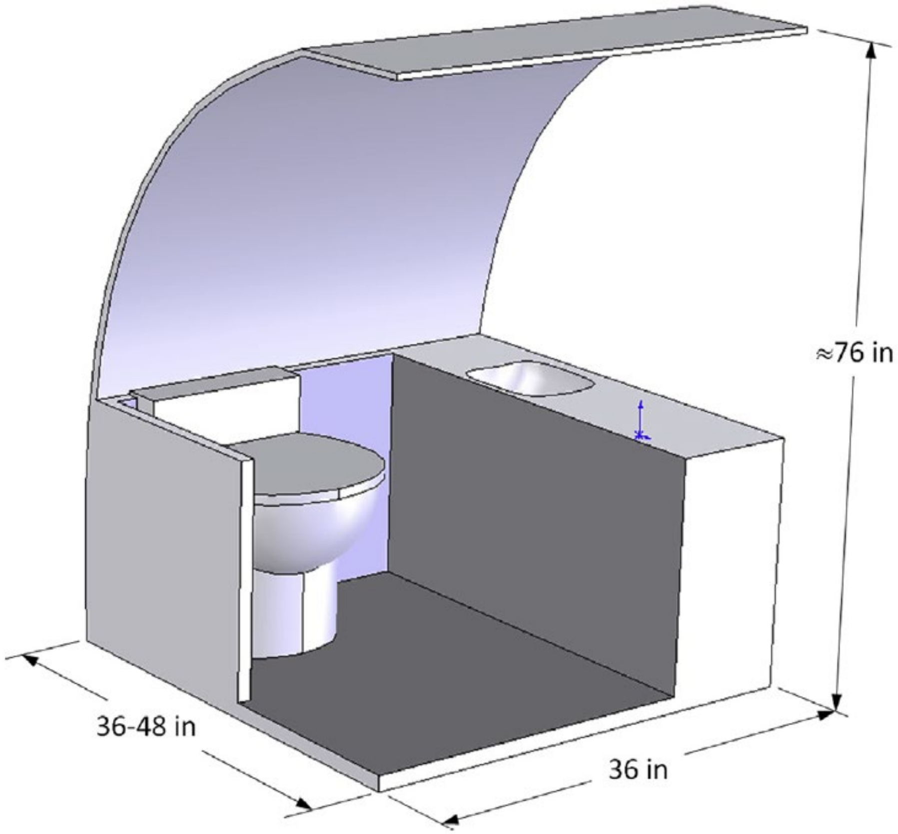
Passenger compartment - configuration

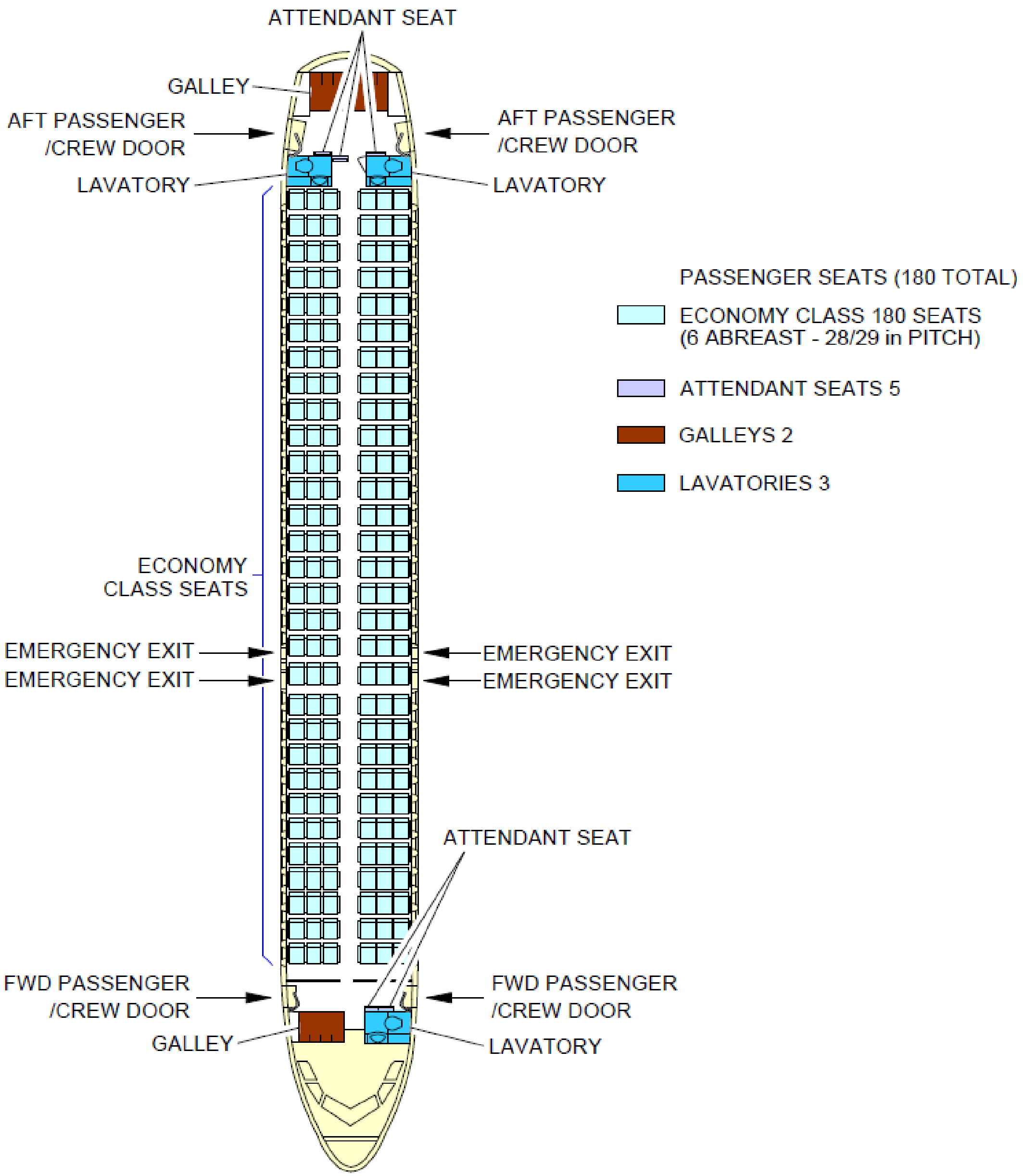
- Number of classes and allowances per class.
- Space for galley and toilets.
- Escape provisions.
- Good idea to look into existing configurations.
- For long range you need space for crew quarters



Passenger compartment – allowances SMR

- Use manufacture, airline data.
  - https://www.boeing.com/commercial/airports/plan_manuals.page
  - https://www.airbus.com/en/airport-operations-and-technical-data/aircraft-characteristics





Passenger compartment – allowances LR

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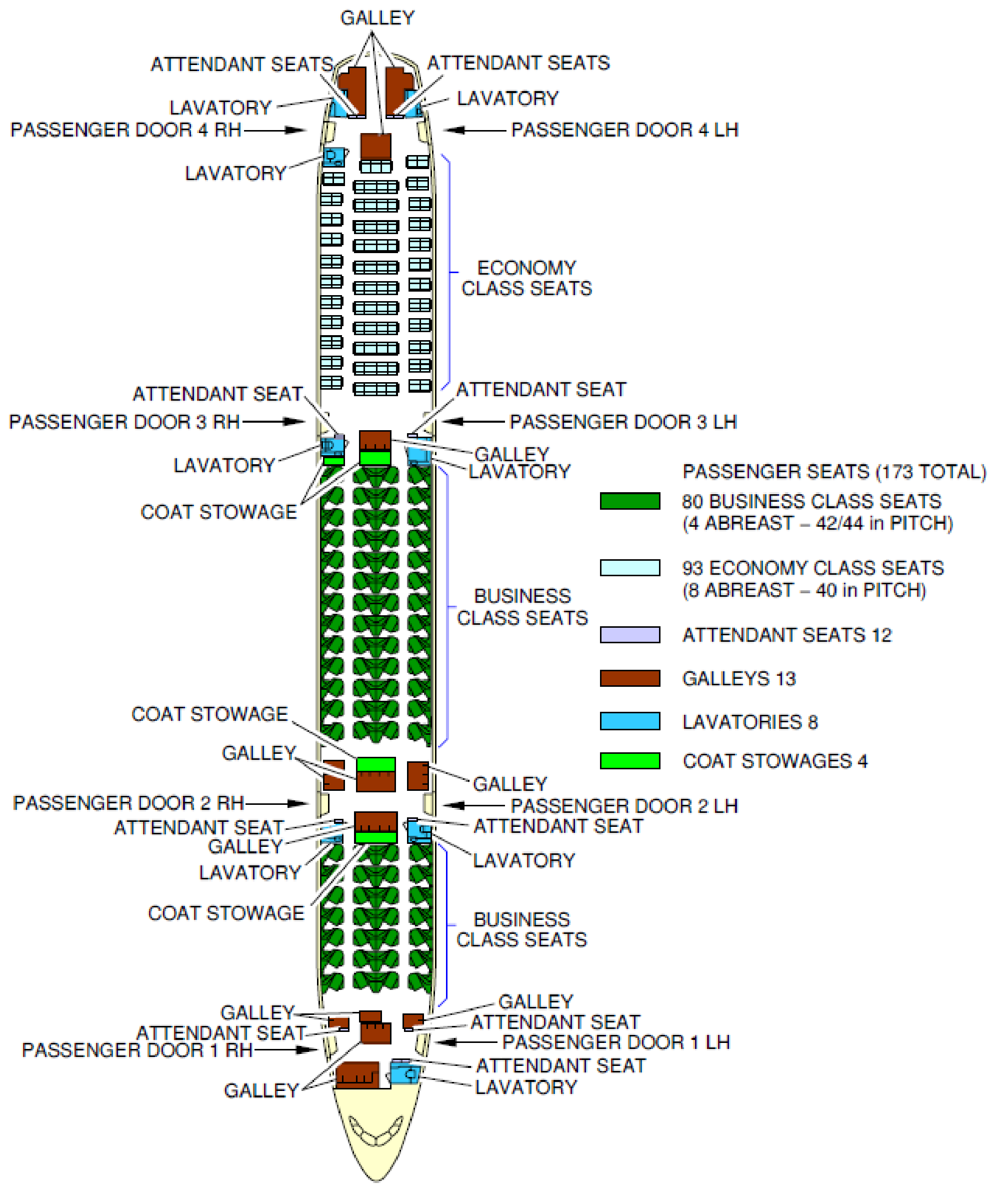
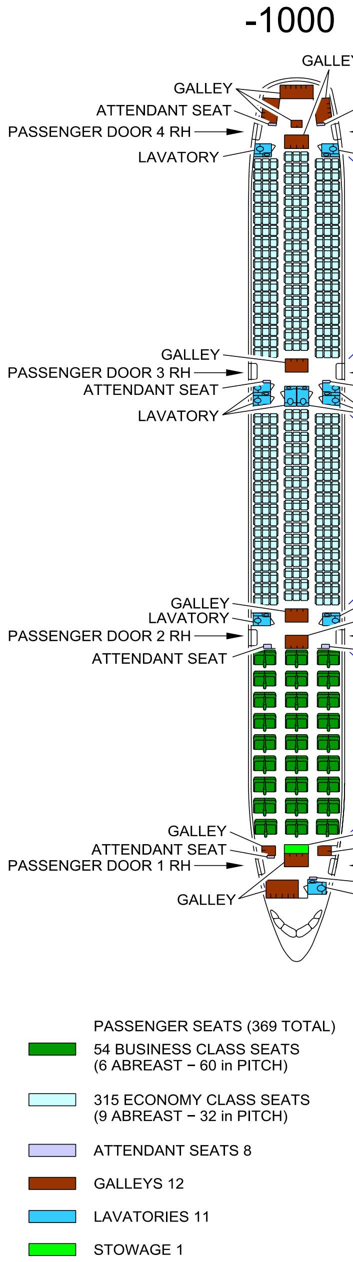
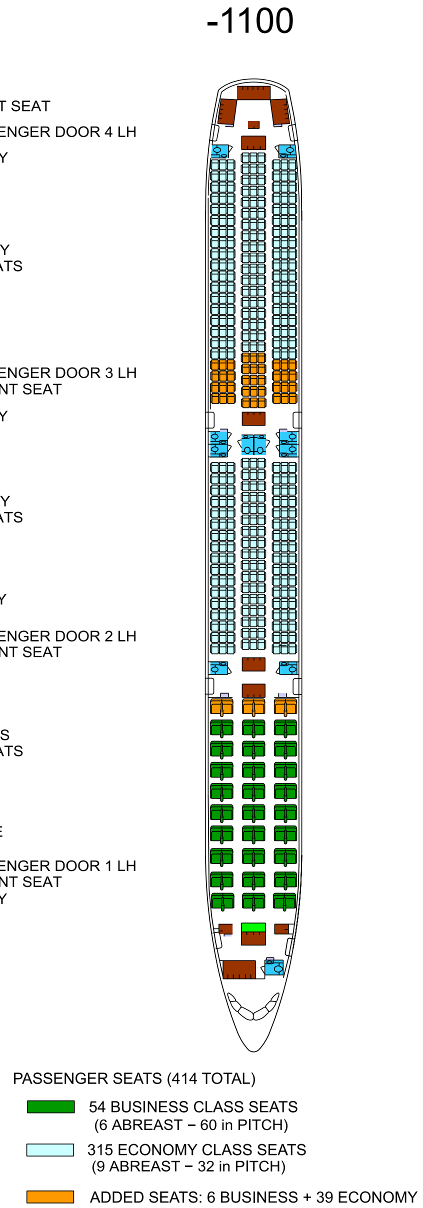
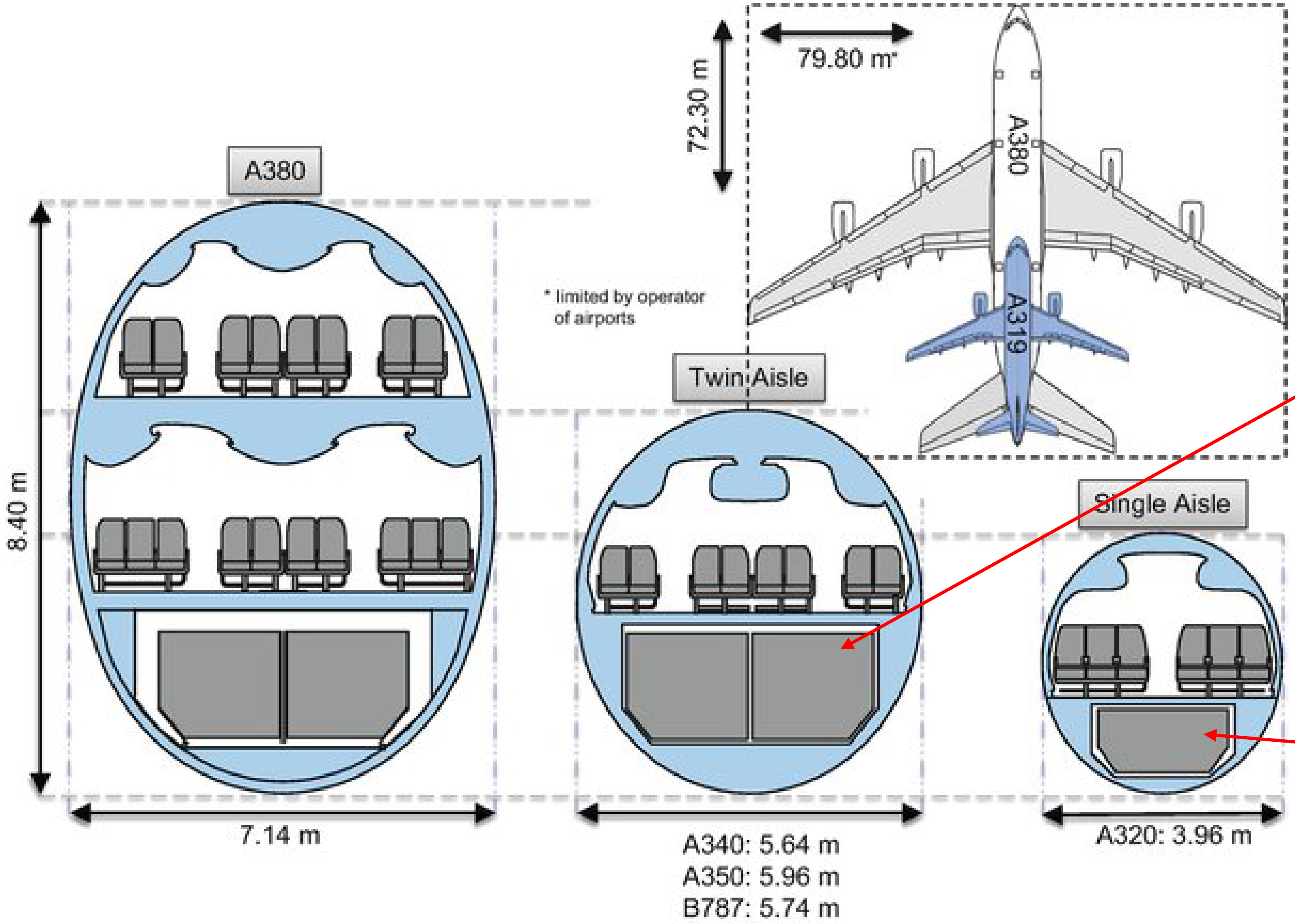
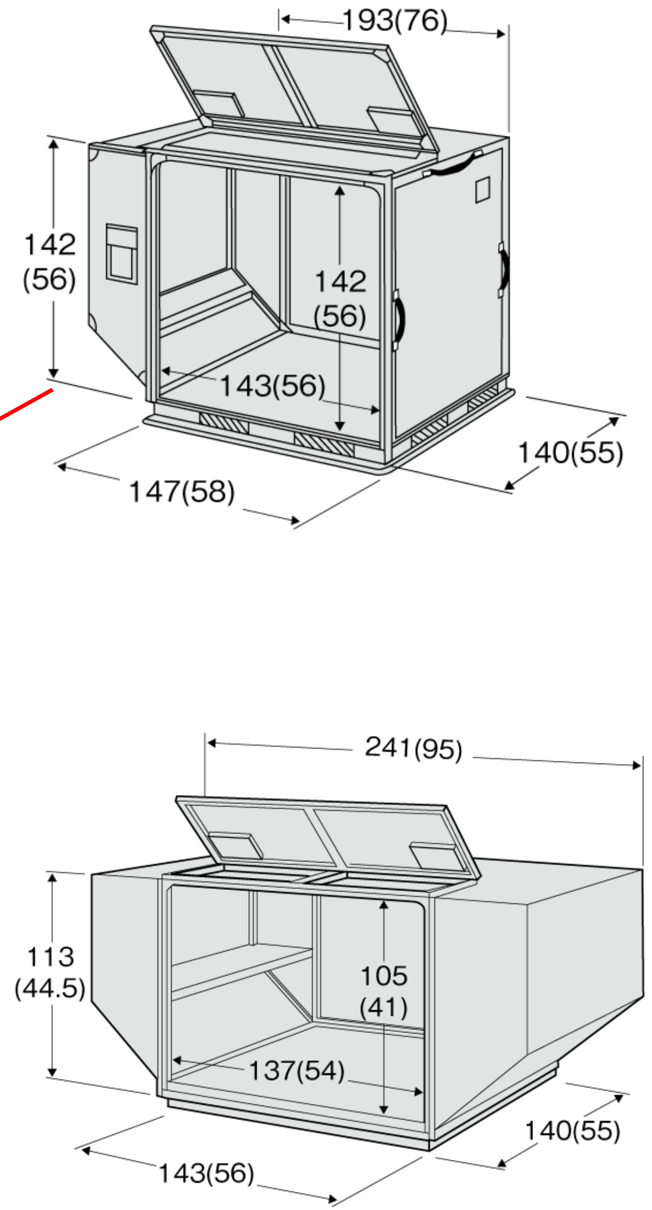




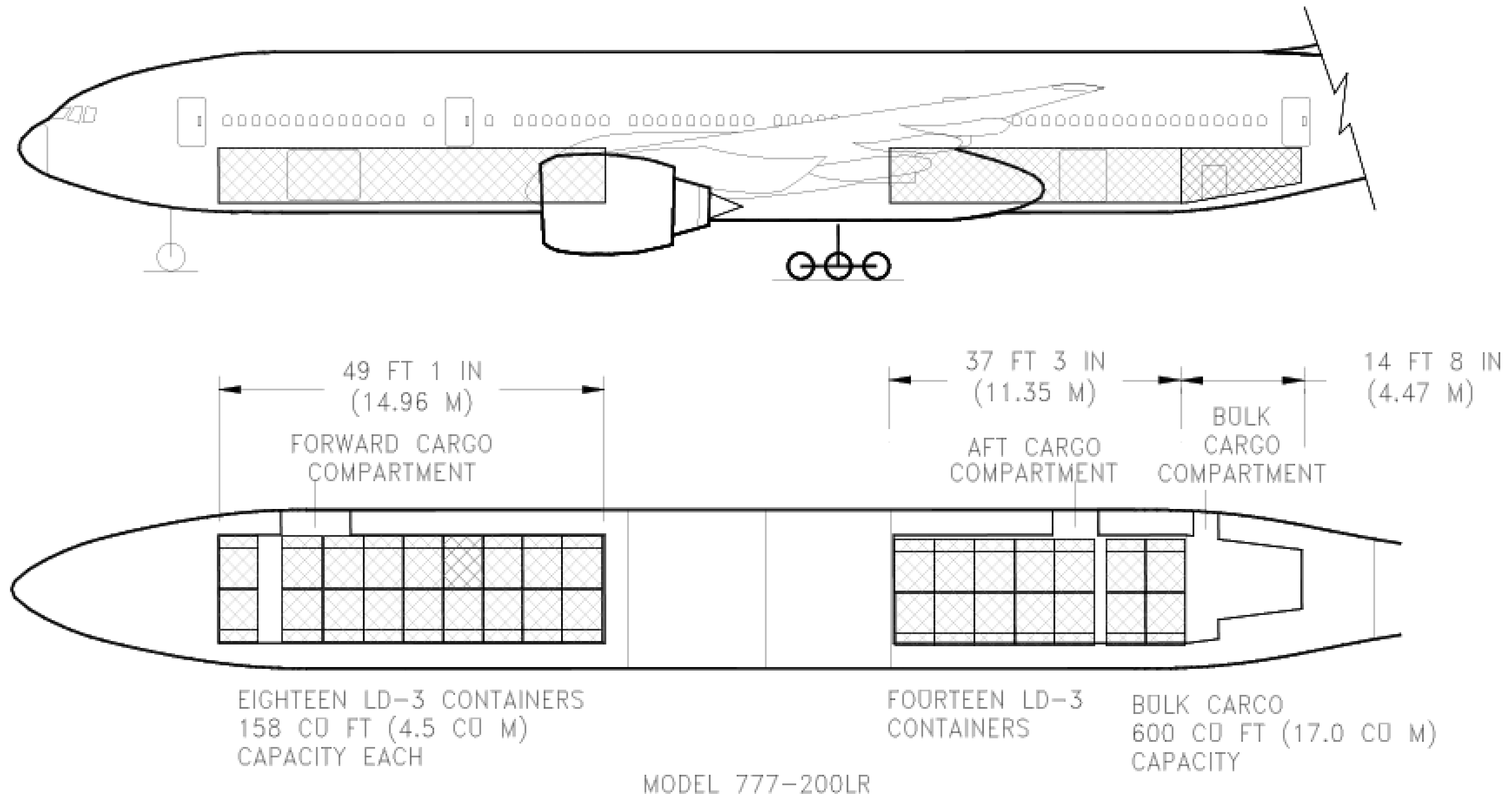


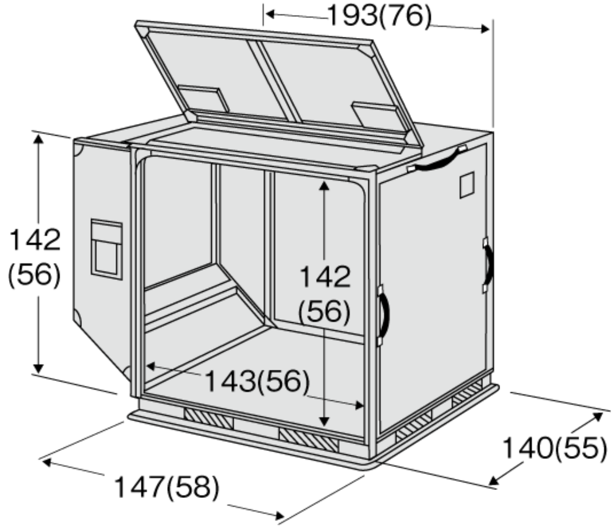
Cabin and cargo – Cross section

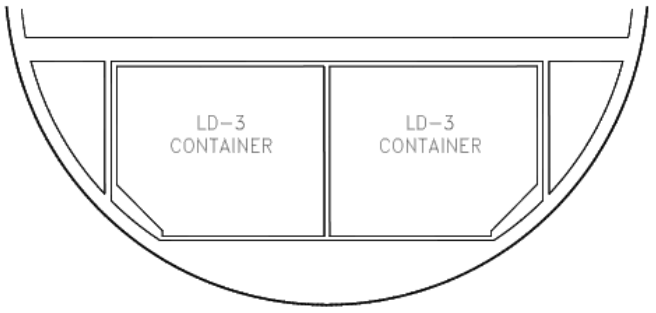




Cabin and cargo – Cargo Containers – 777-200LR

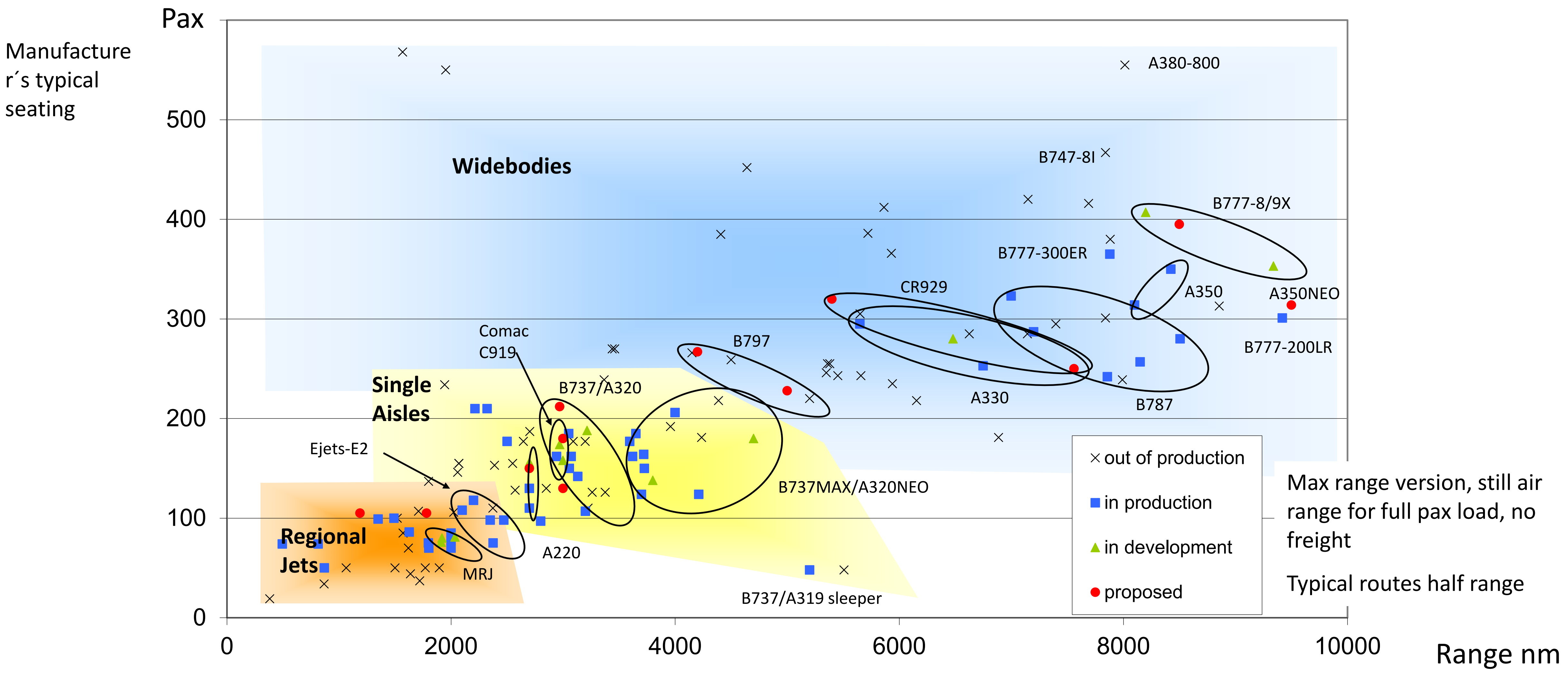






CROSS-SECTION

Capacity-range of aircraft



Max range version, still air range for full pax load, no freight

Typical routes half range

Landing gear

Landing gear - arrangements

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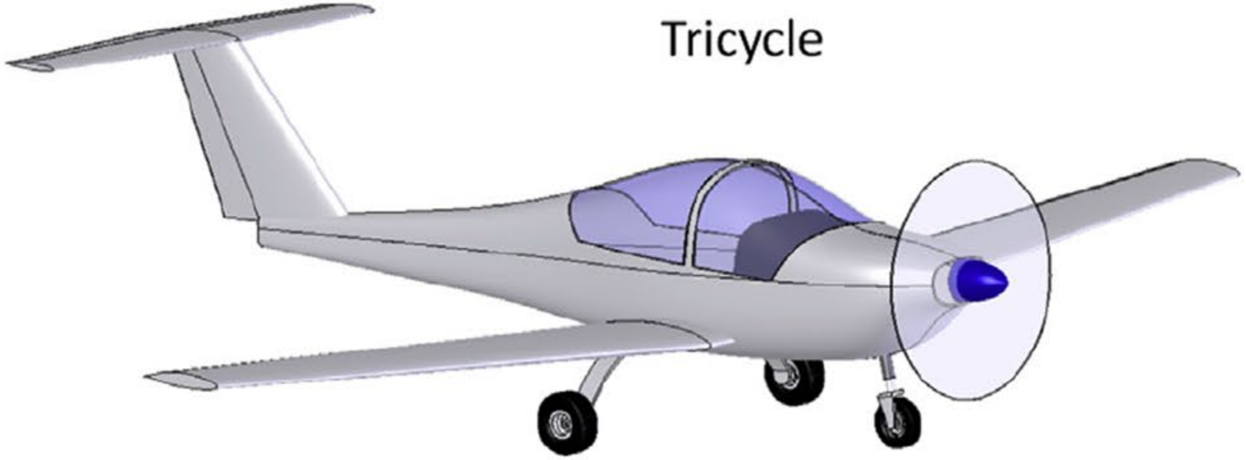
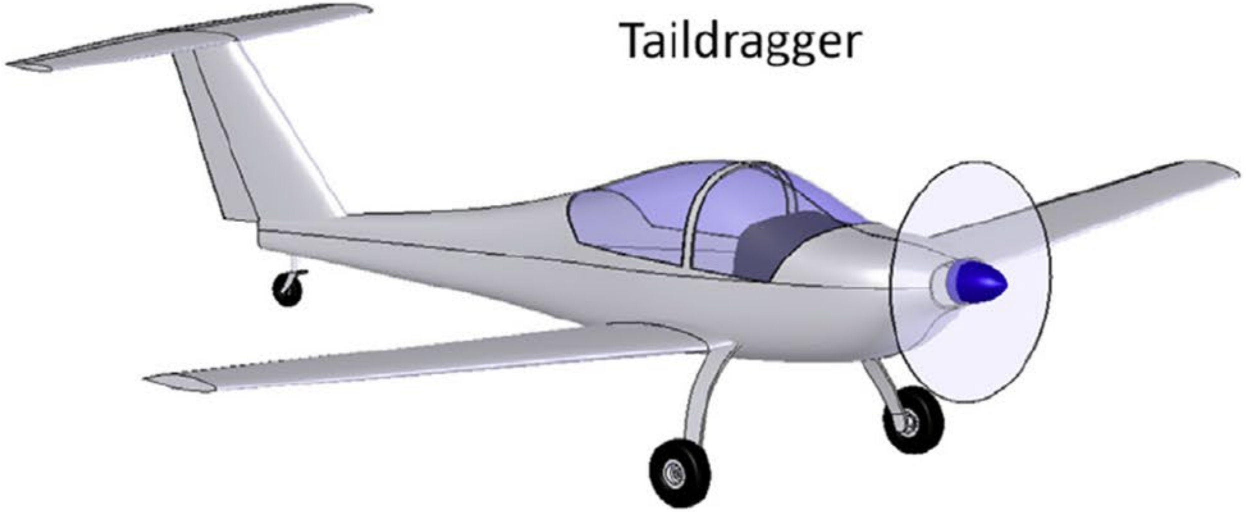
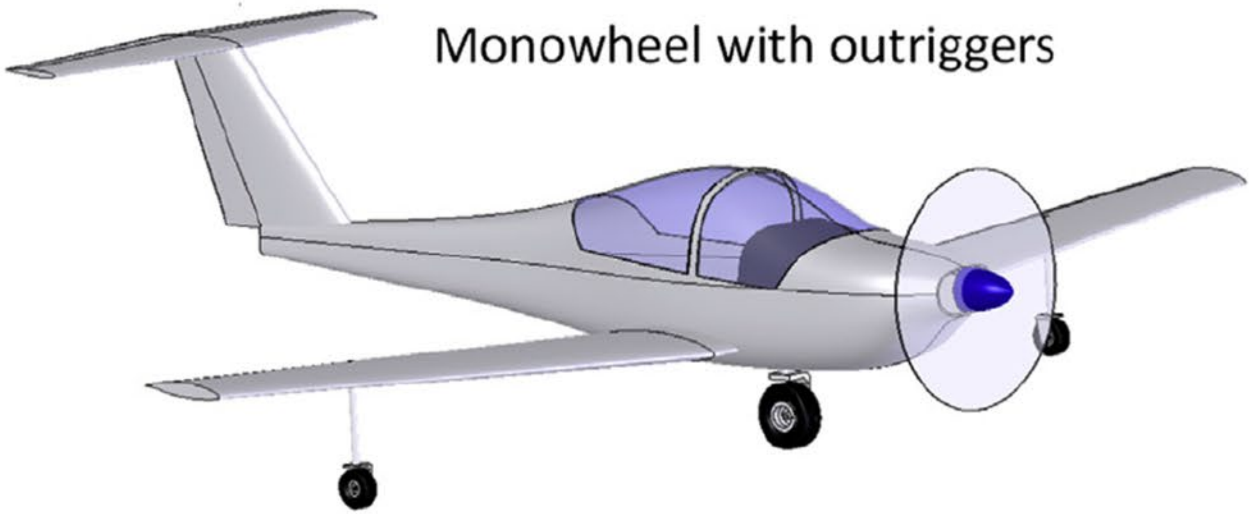
Tricycle



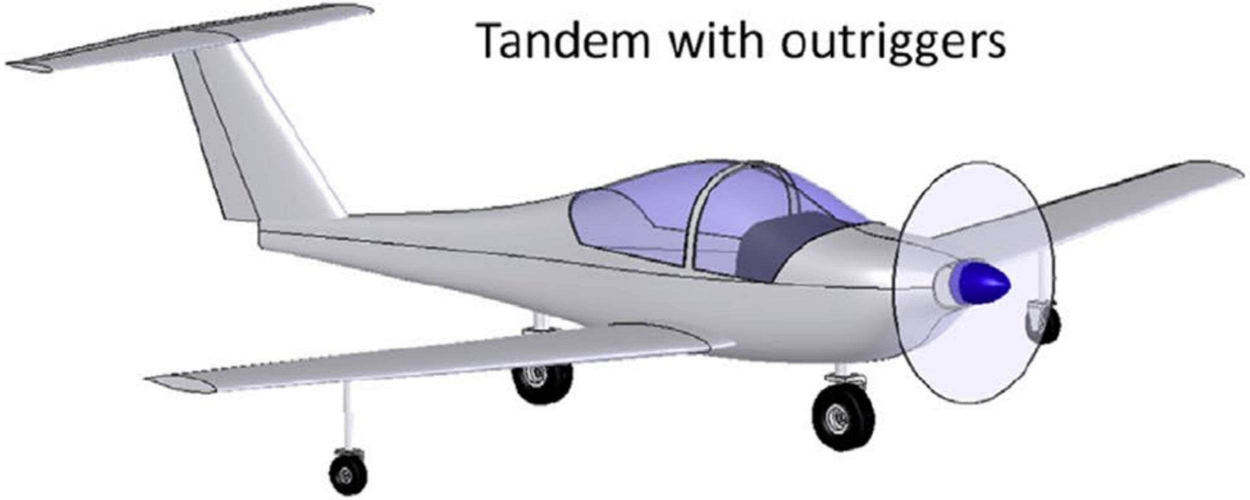
Taildragger



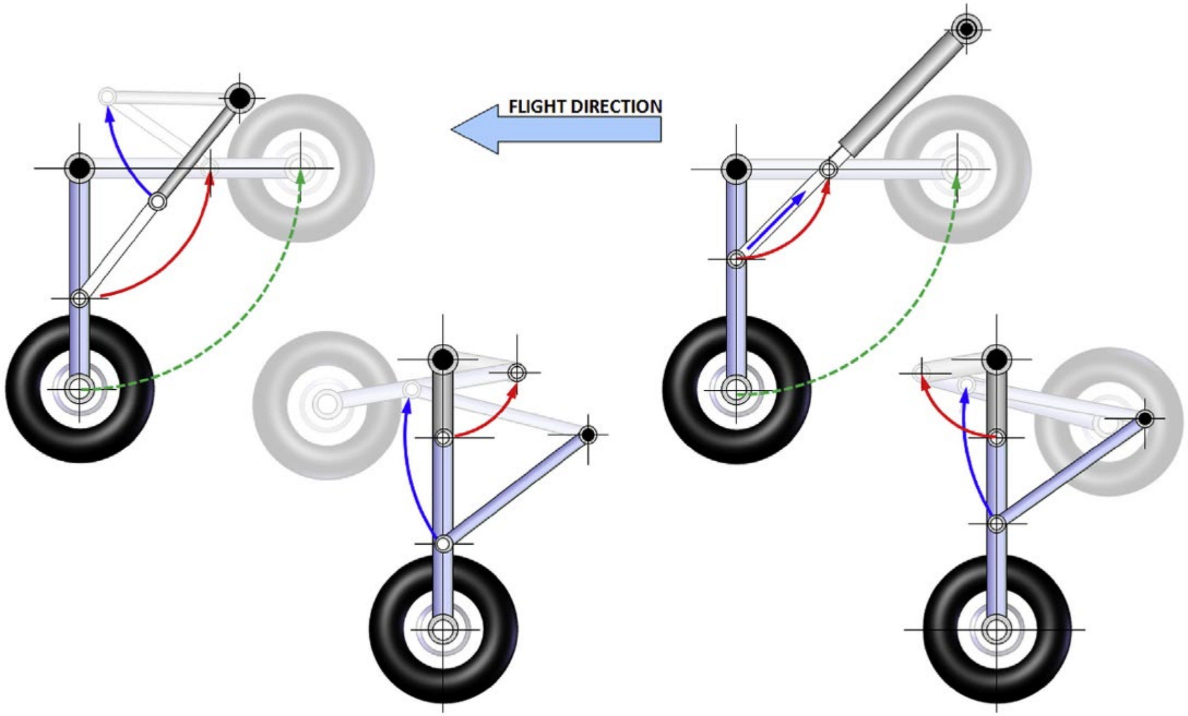
Monowheel with outriggers

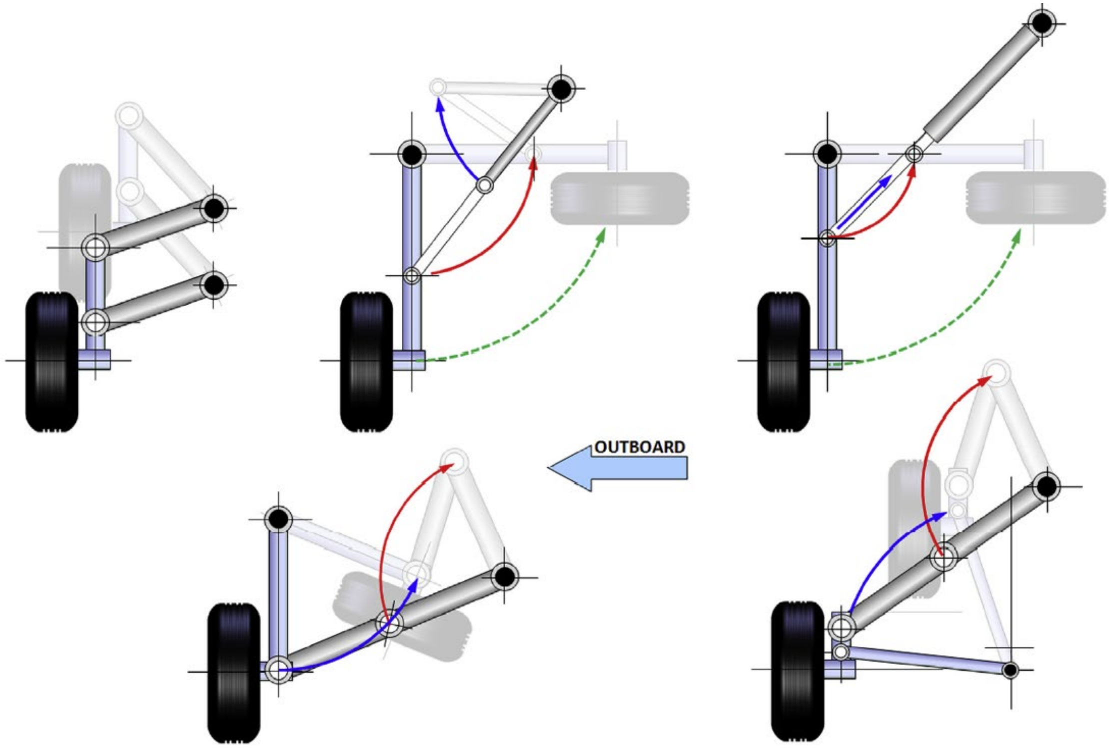


Tandem with outriggers



Landing gear – Retraction (make sure it fits!)



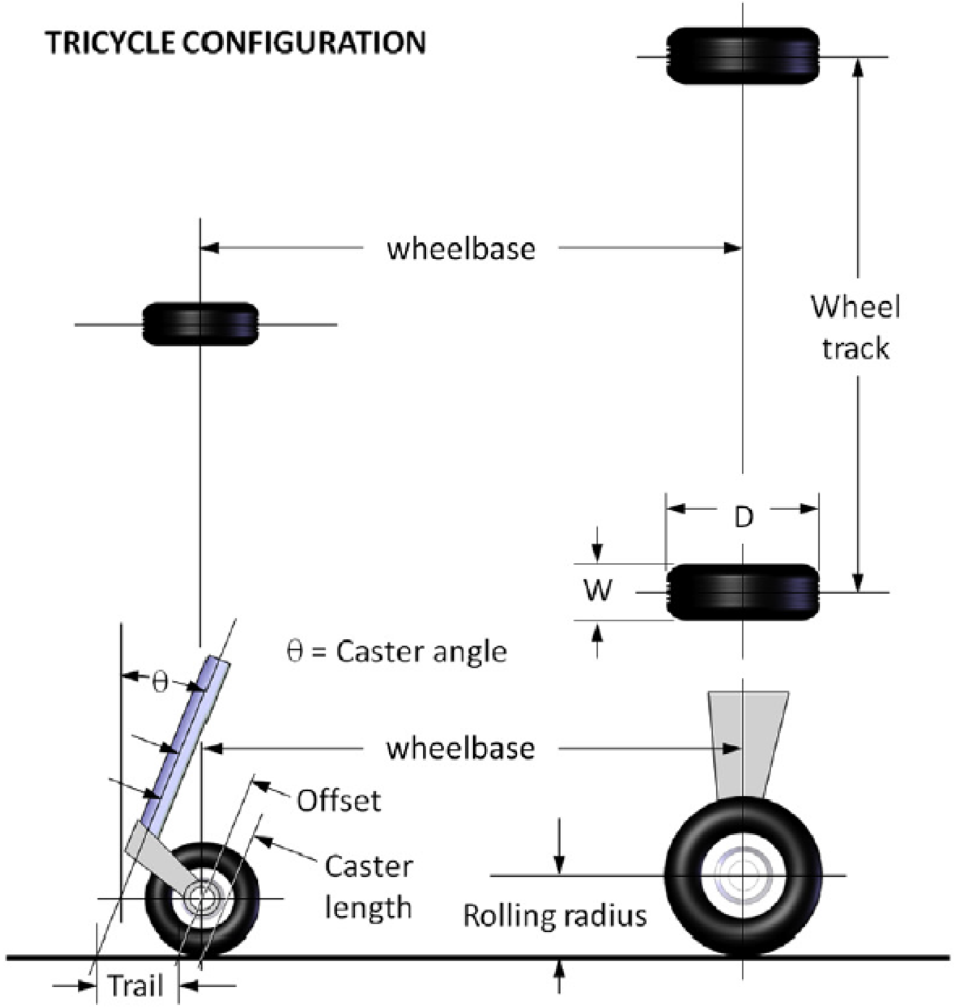


Landing gear – tire sizing

- For sizing the main tires use statistical data (Tab 11.1).

$AW^{B}_{on\ wheel}$				
	Diameter		Width	
	A	B	A	B
General aviation	5.1	0.349	2.3	0.312
Buisness twin	8.3	0.251	3.5	0.216
Transport	5.3	0.315	0.39	0.480

  - The main tires carry about 90% of the weight
  - Nose tires carry at least 10%, but no more than 20%.
- Nose wheels vary between 60% - 100% of main



Landing gear – tire sizing

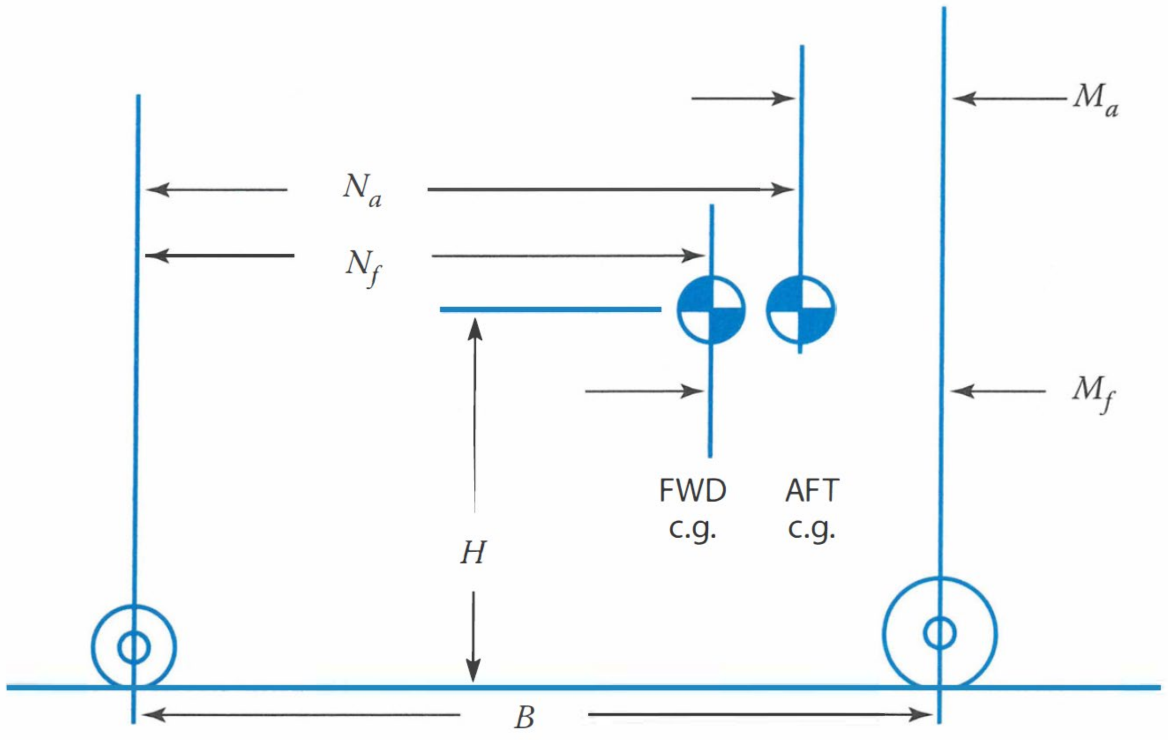
- In the final layout choose from a catalog (Tab. 11.2).
  - Requires a CG envelope
  - Make sure the size can withstand the static and dynamic loads.
  - FAR25 requires a 7% margin in the calculated loads

$$\text{Max Static load} = W_0 \frac{N_a}{B} g_0 \ \ [N]$$

$$(\text{Max Static load})_{\text{nose}} = W_0 \frac{M_f}{B} g_0 \ [N]$$

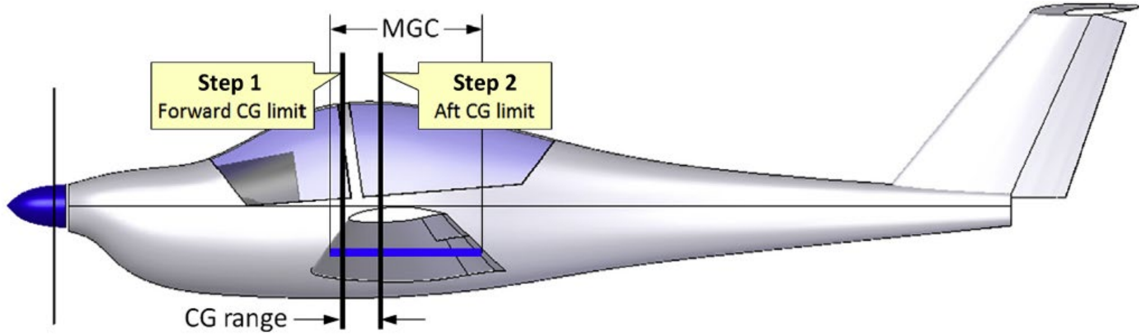
$$(\text{Min Static load})_{\text{nose}} = W_0 \frac{M_a}{B} g_0 \ [N]$$

$$(\text{Dynamic brake load})_{\text{nose}} = \frac{3HW_0}{B} \ [N]$$

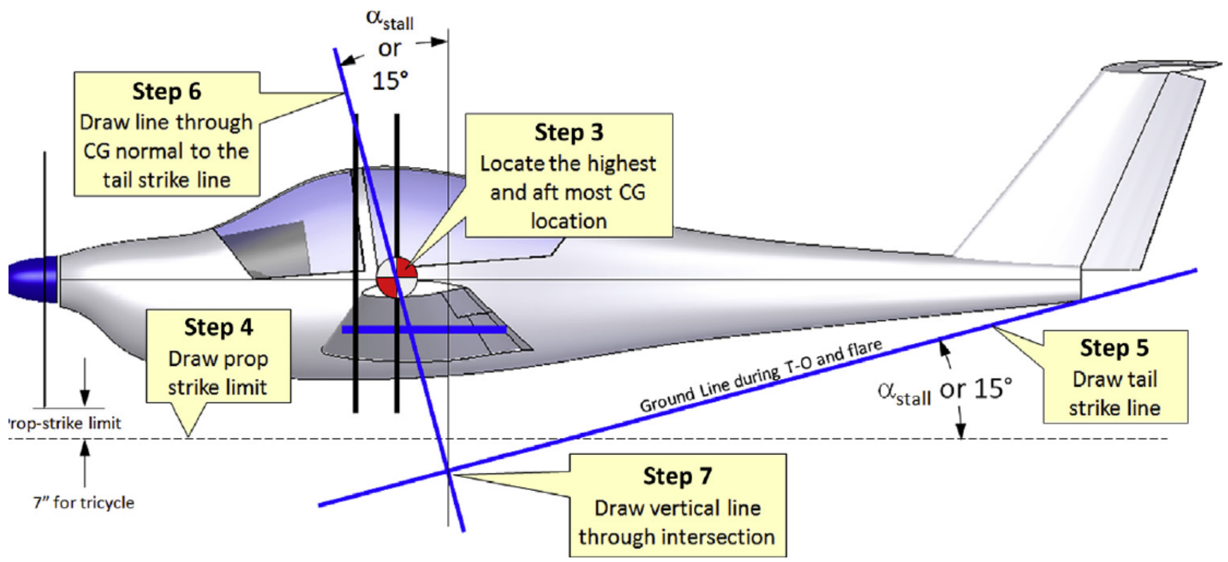


Geometrical layout of a tricycle landing gear

- Step 1 & Step 2: Draw forward and aft CG limit



- Step 3: Determine the highest vertical CG location



(requires a complete CG envelope)

Geometrical layout of a tricycle landing gear

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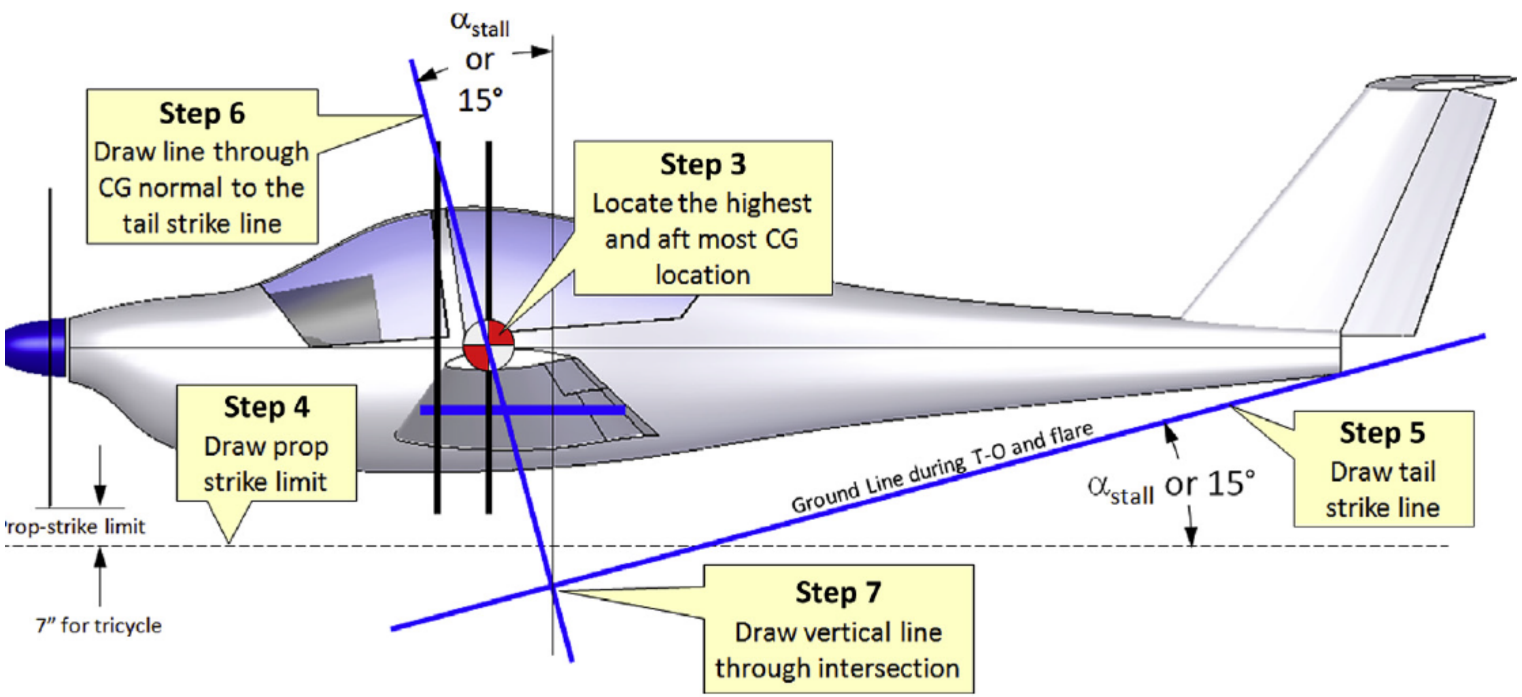
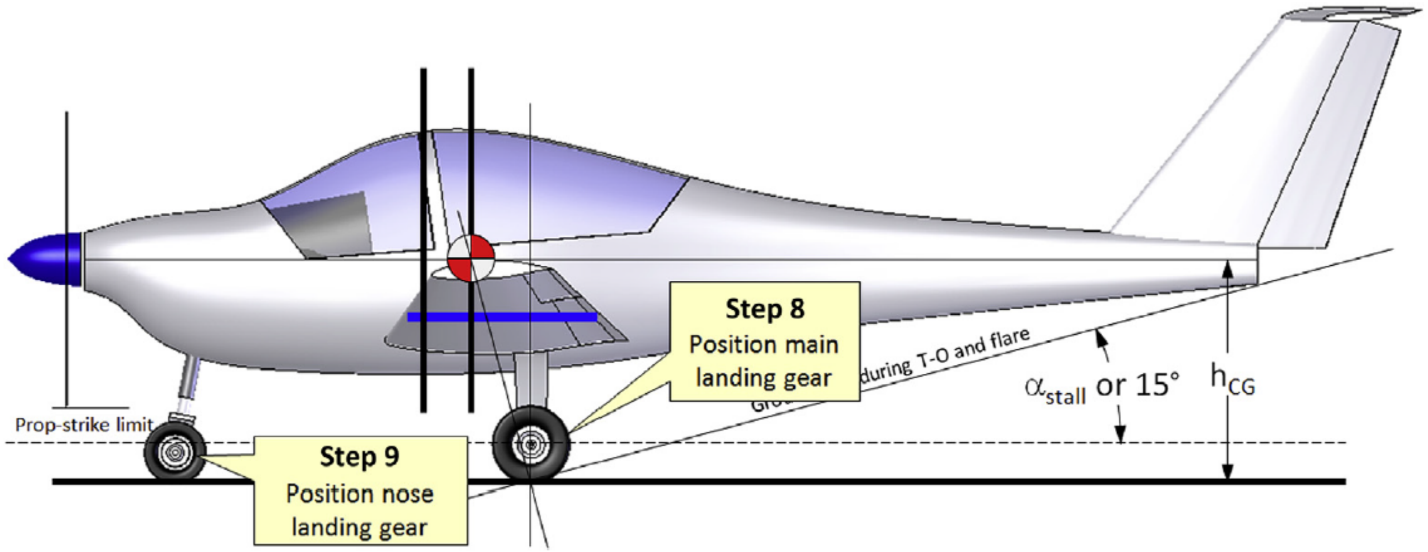


- Step 4: Draw the prop-strike limit as shown.
- Step 5: Draw the tail strike line (ensuring it is at the stall angle)
- Step 6: Draw a line trough the CG perpendicular to the tail strike line => contact point of tire at static deflection of 1g load.

Geometrical layout of a tricycle landing gear

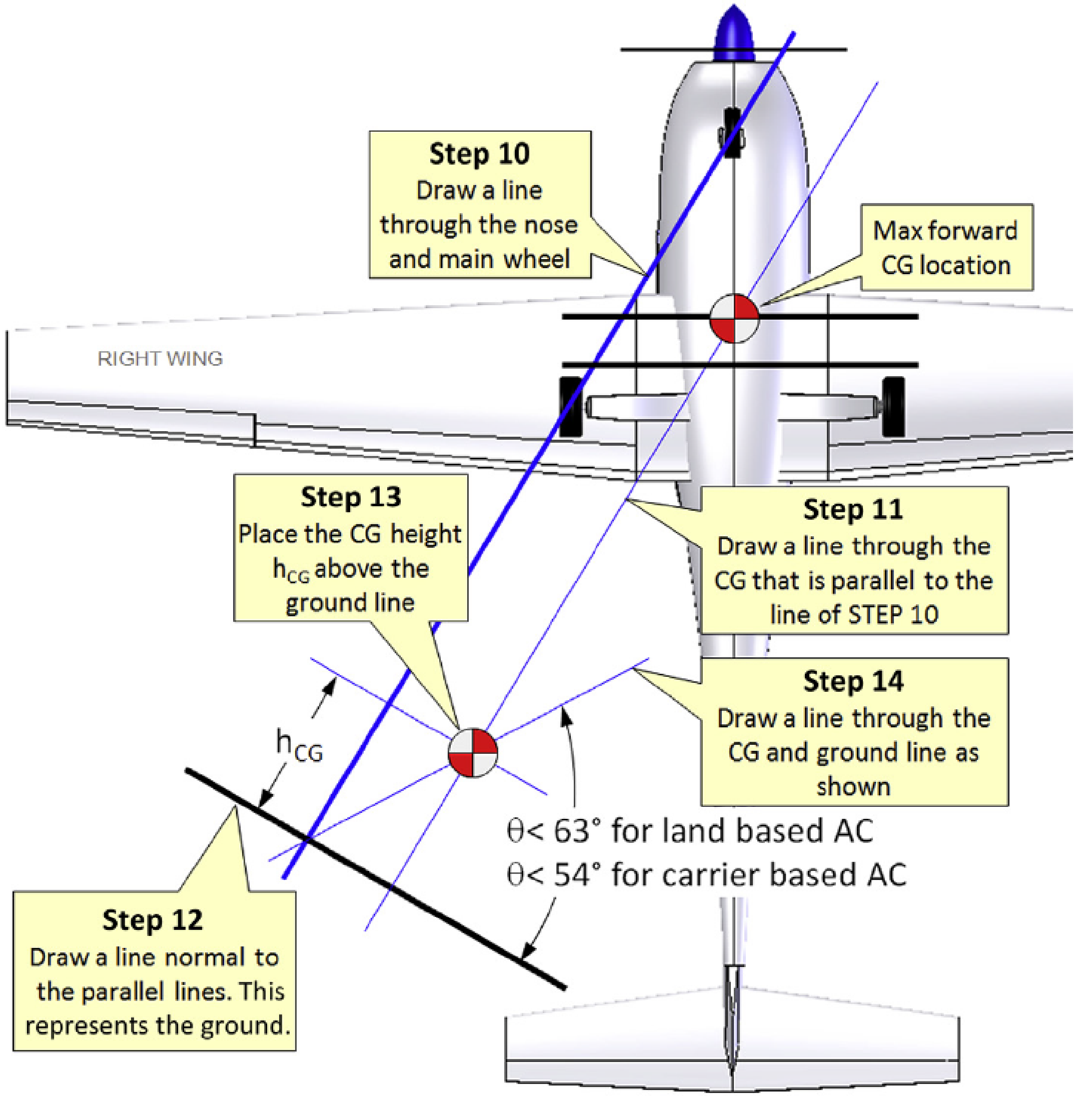


- Step 8: Position the main landing gear => contact point intersection point

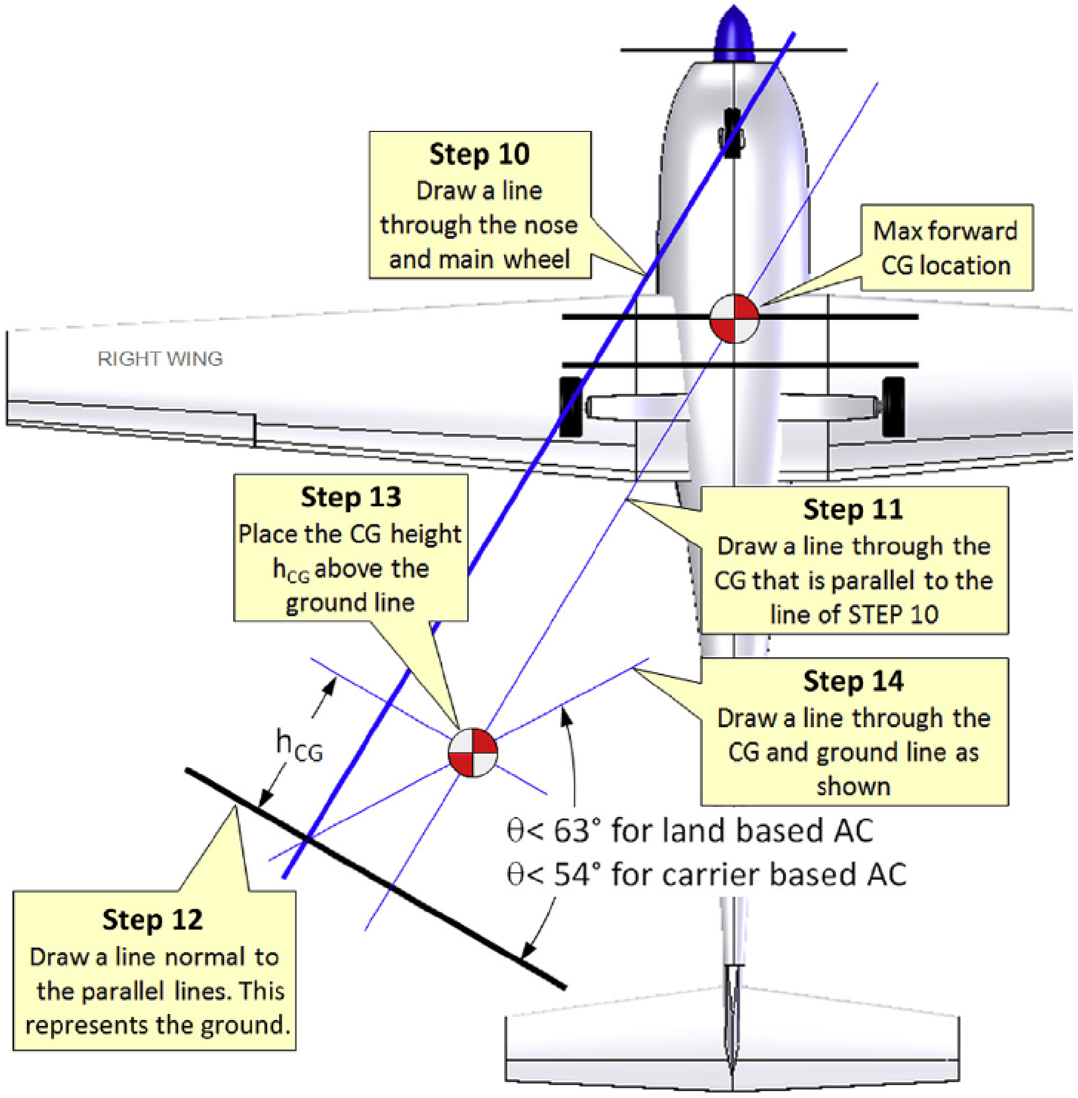
- Step 9: Position the nose gear to carry:
  - No more than 20% of the weight when the CG is at the forward limit
  - No less than 10% of the weight when the CG is at aft limit

Geometrical layout of a tricycle landing gear

- Step 10: Draw a line through the center of the contact points.
- Step 11: Draw a line through the CG parallel to the line of step 10.
- Step 12: Draw a line (”ground”) perpendicular to the lines drawn in 10 and 11.
- Step 13: place the CG distance h_{CG} above the ground line and along the line of step 11.



- Step 14: measure the overturn angle. If larger then permited:
  - The wheel track can be increase.
  - The height of landing gear can be increased.
  - The spacing between nose and main landing gear can be increase.



Thank You!

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